Using Accounting Earnings and Aggregate Economic Indicators to Estimate Firm-level Systematic Risk*

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Abstract

We revisit the literature on using accounting earnings to estimate firm-level systematic risk, using macroeconomic indicators rather than listed-firm indexes to measure aggregate risk. Conventional listed-firm indexes reflect an unrepresentative subset of aggregate assets and thus are expected to substantially mis-measure aggregate and systematic risk (Roll, 1977). That choice dictates using earnings rather than returns to measure firm-level outcomes. Earnings and macroeconomic indicators both are primarily realized annual outcomes and thus are better aligned in time than forward-looking returns for capturing the contemporaneous co-movements that underlie systematic risk. Our macroeconomic indicators are chosen to reflect shocks to aggregate supply and demand, providing a parsimonious model incorporating the two fundamental determinants of aggregate risk. We find that firms' earnings-based sensitivities (betas) to aggregate supply and demand shocks are negatively correlated, and explain the cross-section of returns better than conventional "index" betas. They are correlated with firm characteristics employed in empirical asset pricing models, and explain one quarter of the explanatory power of those characteristics.

JEL classification: G12, M41

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1 Introduction

We revisit an extensive literature, from Ball and Brown (1969) and Beaver, Kettler, and Scholes (1970) to Penman and Zhang (2020), on the information about firm-level systematic risk contained in accounting earnings. Systematic risk here is defined as firm-level sensitivity to undiversifiable aggregate economic shocks. We depart from that literature by measuring aggregate outcomes using macroeconomic indicators rather than conventional indexes of publicly listed firms' earnings. Our premise is that the macroeconomic indicators we select reflect a substantially wider set of economic outcomes than indexes that cover only publicly-listed firms. Shocks to these indicators therefore provide better measures of systematic risk.

It is well-known that firms' earnings comove with aggregate economic indicators, but the firm-level determinants of earnings co-movements and hence of systematic risk are less well understood. We show that systematic earnings risk originates in different economic forces across firms. Some firms' earnings are more sensitive to aggregate productivity shocks, other firms' earnings are more sensitive to aggregate wealth shocks, and the sensitivities are negatively correlated. To validate the results, we show that when differences in where risk originates are taken into account, earnings-based estimates of systematic risk help explain the cross-section of average stock returns. Furthermore, they partially subsume the explanatory power of firm characteristics commonly employed in empirical asset pricing models, suggesting that at least part of the ability of these characteristics to predict returns originates in their correlation with firms' systematic risks.

An important issue in accounting is the manner and extent to which financial statements contain information about firm risk. It has been documented since Brown and Ball (1967) that firms' accounting earnings possess significant systematic components; in other words, firms' profits comove significantly due to aggregate economic forces. While we know from the literature that accounting earnings have significant systematic components (see also Fama and French, 1995; and Ball, Sadka, and Sadka, 2009), we have little insight into the nature of the macroeconomic forces underlying earnings co-movements, how those forces differ across firms, and whether they are priced. We attempt to identify sources of risk information in

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earnings based on their sensitivities to aggregate shocks.

To avoid an *ad hoc* selection of macroeconomic indicators and data mining, we make a parsimonious choice based on the two most fundamental forces in economics that affect firms: shocks to aggregate supply and shocks to aggregate demand. We expect firms' earnings are exposed to both. The macroeconomic indicators we select to proxy for aggregate shocks are changes in Total Factor Productivity (TFP) and changes in aggregate wealth. We employ these indicators, rather than indexes of public firms' earnings, to measure aggregate risk.

We conjecture that firms have heterogeneous production functions and heterogeneous products or services, so they differ in whether they are affected more by aggregate supply shocks or by aggregate demand shocks. We therefore estimate two instruments for firm-level systematic risk: earnings sensitivity to changes in TFP (which we refer to as a supply beta); and earnings sensitivity to changes in total household wealth (demand beta). Although ΔTFP and ΔWealth are positively correlated, we find that firms' supply betas and demand betas are negatively correlated. For example, a low-cost manufacturer's production function might be more dependent on aggregate TFP changes, while a luxury brand's demand might be more sensitive to changes in wealth. We conclude that estimating firms' earnings sensitivities to proxies for both aggregate demand and aggregate supply shocks potentially enhances our understanding and measurement of firms' systematic risks.

We also conjecture that shocks to indicators of macroeconomic activities provide better measures of aggregate risk than changes in conventional indexes based only on public-firm equity claims (represented either by earnings or returns). Public-firm equity claims are a minor and non-random subset of the assets into which households are diversified. Indexes based on public-firm equity claims exclude outcomes on risky public-firm debt, all debt and equity claims on private firms, human capital, stocks of consumer durables such as housing and autos, not-for-profit institutions such as universities, hospitals and charities, and public assets such as ports, roads and airfields. All such assets contribute to aggregate macroeconomic outcomes and hence to systematic risk. As Roll (1977) observes, valid estimates of systematic risk require a measure of aggregate undiversifiable risk that is a mean-variance efficient portfolio; and mis-measuring aggregate risk is equivalent to mis-measuring firm-level

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sensitivity to aggregate risk.[1] We refer to firms' earnings sensitivities to variation in an index of listed firms' earnings as index betas, and challenge using index beta as a single measure of firm systematic risk. We do not propose that the macroeconomic indicators we use are perfect in resolving this measurement problem; rather, we offer them as capturing outcomes on a wider range of assets than conventional indexes, and hence as potentially providing better firm-level measures of systematic risk.

Given our use of annual macroeconomic indicators to measure aggregate outcomes, accounting earnings offer several advantages over stock returns when estimating firm-level systematic risk. The major advantage is that earnings and macroeconomic indicators both are primarily realized past outcomes over the same period. Relative to stock returns, earnings are primarily (though not exclusively) backward-looking, hence prices lead earnings.[2] Accounting rules incorporate some estimates of future cash flows in earnings via accounting accruals, such as booking credit sales as revenue in anticipation of receiving a future cash settlement, or booking asset write-downs when the future cash flows they are expected to generate fall. Nevertheless, much of accounting earnings is created by recording (counting) past realized cash flows. Consequently, stock returns incorporate considerably more information than current earnings about shocks to expected future cash flows.[3] Because returns are more forward-looking than earnings, they are not as well aligned with the macroeconomic variables. Ironically, the fact that prices lead earnings means that returns contain more information than earnings, and in that sense they are superior, but that is what makes returns inferior for the present task. In comparison, annual earnings are better aligned in time with annual measures of aggregate economic outcomes and hence are more likely to capture the contemporaneous co-movements that underlie systematic risk.

Several returns-based studies attempt to address the misalignment issue by using forward-looking variables (e.g., Hansen, Heaton, and Li, 2008; Malloy, Moskowitz, and Vissing-Jørgensen, 2009), but it is difficult or impossible to find expectational variables that match

[1] See also Mayers (1972).

[2] The phenomenon of prices leading returns was first documented by Ball and Brown (1968) and measured more precisely by Ball and Shivakumar (2008).

[3] A formal model of how earnings and returns differ in incorporating forward-looking information is provided by Ball, Kothari, and Nikolaev (2013, Section 2).

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the unobservable and likely lengthy horizon of stock prices.[4] When using listed-firm indexes of accounting earnings to estimate firm-level systematic risk, Da and Warachka (2009) and Ellahie (2020) specify the earnings variable as revisions in analyst earnings forecasts. However, analyst earnings forecasts have limited horizons, and even if made for a one-year horizon they do not perform appreciably better than a simple random walk model (Bradshaw, Drake, Myers, and Myers, 2012), which defaults to our specification of firm-level changes in earnings. In addition, requiring forecast data would substantially restrict the sample to firms covered by analysts and to the period covered by IBES. Using realized earnings extends our sample period from 38 to 55 years and increases the average number of firms per year from approximately 3800 to 5700.

A further advantage of using earnings is that, relative to estimating the principal components of earnings and returns (Ball, Sadka, and Sadka, 2009) or estimating stock returns on factor-mimicking portfolios of unobservable factors (e.g., Fama and French, 2015), our macroeconomic indicators are grounded in accounting and economic constructs. Focusing on productivity shocks and wealth shocks allows us to investigate the economic nature of systematic risk, as distinct from observed empirical correlations with returns. Stock returns also reflect unobservable investor risk preferences and behavioral biases. For example, some studies consider the relation between book-to-market and future returns as a rational risk-return relation (e.g., Fama and French, 1993) but others consider the relation as resulting from a behavioral bias (e.g., Lakonishok, Shleifer, and Vishny, 1994). In contrast, accounting earnings and macroeconomic indicators are mainly realizations of past economic transactions, and are not affected by investor horizons or behavioral biases. For example, a firm's earnings sensitivity to productivity shocks reflects economic fundamentals, while high return sensitivity to returns on a factor mimicking portfolio may not have such a clear economic interpretation.

Using earnings to estimate risk has disadvantages as well. Earnings are based on accounting measurement rules whose effects can differ across firms and across time. In addition, bottom line net income contains earnings components that are more transitory or

[4]Kamara, Korajczyk, Lou, and Sadka (2016) document that asset risk depends on investment horizon.

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discretionary, such as gains and losses on discontinued operations. These components add noise to the earnings variable, so we use operating income—which excludes many of these components—as the earnings variable (Novy-Marx, 2013; Ball, Gerakos, Linnainmaa, and Nikolaev, 2015, 2020). Another disadvantage of using annual earnings is that it is substantially left-skewed. Finally, annual earnings provide fewer time-series observations than monthly returns, so imposing even a modest minimum on the number of time series observations encounters substantial survivorship bias. To reduce noise, skew and survivorship bias, we annually sort firms into portfolios to estimate systematic risk, the trade-off being a reduction in power.

We first document significant correlations between changes in aggregate earnings and changes in both aggregate wealth and TFP, validating the latter as capturing systematic component of earnings. We then study the relation between firms' earnings-based supply and demand betas and four empirical firm characteristics that are known to predict future stock returns: book-to-market, return momentum, size, and asset growth. We find that earnings-based supply and demand betas both increase monotonically in book-to-market and decrease monotonically in firm size and asset growth. Both betas increase in return momentum, but the pattern is not monotonic. If the earnings-based supply and demand betas are indeed capturing the CAPM concept of systematic risk, then these results support our conjecture that at least part of the ability of firm characteristics commonly used to predict returns originates in their correlation with systematic risk.

We also find that TFP and wealth betas are negatively correlated. Some intuition for this result is provided by comparing luxury vacation resorts with pharmaceutical companies. Risk in vacation resorts is likely to originate more in aggregate wealth changes than in productivity changes, whereas risk in pharmaceuticals is likely to originate more in technology shocks. This result illustrates a primary advantage of using macroeconomic indicators to measure aggregate risk and using accounting earnings to measure firm-level exposure to aggregate risk: the procedure lends itself to investigating firm or industry level heterogeneity in economic sources of systematic risk.

We next address pricing, using a Fama and MacBeth (1973) procedure to examine

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earnings-based index, supply, and demand betas in the cross section of average stock returns. We initially show that earnings-based index betas, estimated from linear regressions of firm earnings on an index of listed-firm earnings, carry a significant market risk premium, complementing the evidence in Ball, Sadka, and Sadka (2009), Da and Warachka (2009), and Ellahie (2020). Index betas are positively correlated with both supply (TFP) and demand (wealth) betas. More importantly for our thesis that macroeconomic indicators measure aggregate risk more effectively than earnings indexes, we show that both earnings-based supply and demand betas carry a positive and significant risk premium.

Finally, we show that, after controlling for earnings-based supply and demand betas, the collective explanatory power of the four empirical firm characteristics we study (book-to-market, return momentum, size, and asset growth) declines by approximately one quarter. This result is consistent with the conjecture that the predictive ability of these firm characteristics is due at least in part to their correlation with underlying economic sources of systematic risk that are observed with error.

We caution that our results are based on instruments for aggregate risk derived from supply and demand shocks that inevitably are imperfect. We note that monthly firm-level stock returns are highly variable and hence, as in all pricing tests, the explanatory power of our two earnings-based betas is weak at the firm level. We also note that results regarding return momentum portfolios are sensitive to the choice of a deflator. Lastly, we fail to find that firms' sensitivities to aggregate demand or supply shocks are associated with earnings-to-price, post-earnings-announcement-drift, profit margin, or retained earnings to market equity. Overall, we offer our tests as an exploratory investigation into using a combination of accounting earnings and macroeconomic indicators to directly estimate systematic risk and identify its economic sources.[5]

[5]We offer our estimates of earnings-based supply betas and demand betas as instruments for CAPM systematic risk. Both are based on individual firm co-movement with aggregate undiversifiable outcomes. We do not offer them as yet another set of empirical asset pricing factors. Quite the opposite: we report evidence that these admittedly imperfect estimates of CAPM systematic risk displace some of the explanatory power of empirical factors.

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2 Literature on Accounting Earnings and Systematic Risk

An important issue in accounting is the manner and extent to which financial statements contain accounting information about firm risk. A common research question has been whether and how accounting earnings contain information about firm-level systematic risk. Building on the Brown and Ball (1967) finding of a substantial aggregate "market" factor in firms' earnings, Ball and Brown (1969) report a positive correlation between returns-based and earnings-based index (CAPM) betas. Gonedes (1973) replicates this result using different specifications of the earnings variable. Beaver, Kettler, and Scholes (1970), Rosenberg and McKibben (1973) and Beaver and Manegold (1975) conclude that supplementing earnings-based betas with accounting measures of firm characteristics has greater explanatory power than returns-based betas. Thompson (1976) investigates the relation between returns-based betas and betas calculated from a variety of accounting and financial variables, including dividends betas, earnings betas, earnings-multiple betas, leverage betas, and sales revenue betas. Eskew (1979, 116) reports a "small but consistent superiority" of earnings-based betas over returns-based betas when predicting future returns-based betas. Hill and Stone (1980) reach a similar conclusion using earnings betas estimated from both return on assets (operating income scaled by book value of total assets) and return on equity (net income scaled by book value of total equity). In contrast, Elgers (1980) finds that earnings betas offer no improvement over returns betas, though Elgers and Murray (1982) subsequently attribute these differences to researchers choosing different stock indexes (in particular, the S&P500, CRSP value weighted, and CRSP equal weighted indexes). Baran, Lakonishok, and Ofer (1980) report that earnings and returns betas are correlated, and the correlation is largely independent of whether earnings are adjusted for general price level effects (i.e., inflation and deflation). Ismail and Kim (1989) conclude that accounting betas calculated from cash flows are significantly better at explaining returns-based betas than accounting betas calculated from earnings.

Penman and Zhu (2019) connect accounting principles (e.g., clean surplus and earnings

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recognition) with consumption risk and use accounting numbers to construct a pricing factor that explains cross-sectional stock returns. Penman and Zhang (2020) demonstrate that conservative accounting rules do not record earnings on individual transactions until they are approximately zero-beta (i.e., realized in cash or near-cash), though we note that firm-level (as distinct from transaction-level) systematic risk is a property of the uncertain volume and profitability of transactions.

We study the concept of systematic risk that is central to modern portfolio theory (Markowitz,1952; Sharpe, 1964; Lintner, 1965). Our focus on systematic risk does not imply that own-firm risk (risk of the firm or its securities considered in isolation) is not important. Contexts in which own-risk matters include those with non-linear payoffs such as options and risky debt, contexts with high bankruptcy costs, and investors with concentrated portfolios. Nor does our selection of systematic risk to study imply that accounting information plays no role in such contexts. For example, Konchitchki, Luo, Ma, and Wu (2016) examine earnings downside risk, and show that it predicts firms' vulnerabilities to economic downturns and affects their costs of capital. Our study does not address extreme outcomes and treats positive and negative shocks symmetrically. Nor does studying systematic risk imply that idiosyncratic risk (the component of own-firm risk that is not systematic) is unimportant, despite evidence of a recent decline in idiosyncratic risk (Brandt, Brav, Graham, and Kumar, 2010; Bartram, Brown, and Stulz, 2018).

The closest literature to the present project is Ball, Sadka and Sadka (2009), Da and Warachka (2009), Ellahie (2020), and İmrohoroğlu and Tüzel (2014). Ball, Sadka and Sadka (2009) show that the principal components of earnings and returns exhibit substantial canonical correlations, providing completely empirical evidence (i.e., not based on any economic theory) that the sources of earnings risk and returns risk are closely related. They also show that systematic earnings risk, in the form of sensitivity of firm-level returns to the identified earnings components, explains a significant portion of the cross-sectional of stock returns. Our results provide a theory-based equivalent. Da and Warachka (2009) estimate earnings-based index betas from revisions in analyst consensus earnings forecasts, for decile portfolios sorted on book-to-market, size, and long-term return. The estimated betas are increasing

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in book-to-market and decreasing in size and long-term return, and carry a significant risk premium. Ellahie (2020) also estimates index betas using analyst earnings forecasts, and finds that they increase in leverage, sales and earnings volatility, and decrease in size and profitability. Index betas estimated from changes in earnings forecasts explain returns in cross-section and across time better than do equivalent betas estimated from returns. İmrohoroğlu and Tüzel (2014) study the relation between firm-level TFP and stock returns; in Section 7.5 we reconcile our results with theirs.

3 Measuring Aggregate Risk

The research on earnings and systematic risk surveyed in the previous section uses the variability of an index of listed-firm earnings to proxy for aggregate risk. While the ready availability of data for listed firms makes this choice attractive, publicly listed firms would seem unlikely to generate a representative subset of economic activity and risk. One indication of the limitation of public-firm data is that the fraction of aggregate manufacturing sales revenue generated by listed firms is only 18.4%.

A notable omission from public-firm indexes is private firms. According to Nagar, Petroni, and Wolfenzon (2011), the U.S. Census reports approximately seven million corporate tax filers. Lisowsky and Minnis (2020) study approximately 200,000 tax returns of medium-to-large private firms. These firms vastly outnumber public firms in all major industry groups, even when the sample is limited to firms with revenues over $100 million. They collectively invest more than $9 trillion of capital. They appear to be subject to different risks. Thus, Moskowitz and Vissing-Jørgensen (2002) document a correlation of 0.70 between an annual index of public firm returns based on book values of equity and an index of returns on proprietorships and partnerships. This implies that approximately one half of the variance in the two indexes is not common. It would be surprising if public and private asset returns were uncorrelated because public and private firms compete in the same product markets and are affected by similar macroeconomic factors. Equally, it would be surprising if they were perfectly correlated. Moskowitz and Vissing-Jørgensen (2002, 747) conclude "... a

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complete understanding of investor portfolio choice requires private equity holdings to be endogenized."

Moreover, aggregate risk is affected by more than just the performance of equity claims to public and private firms. Aggregate risk also reflects risky debt claims to public and private firm cash flows. It reflects the performance of human capital (firms' earnings and returns are net of labor income), of consumer durables such as housing and autos, of not-for-profit institutions such as universities, hospitals and charities, of public assets such as schools and municipal facilities, and of infrastructure investments such as utilities, ports, roads and airfields. When so many asset categories are excluded, aggregate risk is likely to be measured with substantial error, and consequently firm-level sensitivity to aggregate risk is likely to be substantially mismeasured (Roll, 1977). This observation provides the primary motivation for using macroeconomic indicators–rather than earnings indexes–to represent aggregate economic outcomes.

4 Proxies for Aggregate Shocks Affecting Earnings

We choose shocks to TFP to measure aggregate supply-induced risk. Macroeconomists have long used TFP to gauge national economic productivity and growth (e.g., Solow, 1956; Lucas, 1990; Manuelli and Seshadri, 2014), because it measures the quantity of output per unit of input. Assume an economy produces Y at time t, where Y represents Gross Domestic Product. A Cobb-Douglas production function yields TFP as the term A in the equation $Y(t) = A(t)K(t)^{\alpha}L(t)^{1-\alpha}$, where K is capital invested and L is labor input. In other words, TFP captures the efficiency of production rather than the quantity of inputs.

Productivity shocks imply that firms in aggregate are producing more or less output from a given quantity of resources. Earnings are sales revenues (i.e., outputs) minus costs (i.e., inputs). Aggregate productivity shocks therefore affect firms' sales revenues relative to their costs, and hence they affect aggregate earnings. We expect the sensitivities of individual firms' earnings to changes in aggregate TFP to differ. If a firm's production function is more or less sensitive to aggregate productivity changes (e.g., technological innovation), the firm's

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earnings are likely to comove more or less with the change in TFP, which is why we choose the sensitivity of firm-level earnings to changes in aggregate TFP to proxy for the firm's exposure to aggregate supply risk. It provides a source of measurable variation across firms in systematic earnings risk based on economic fundamentals.

Similarly, we choose changes in total household wealth to proxy for shocks to aggregate demand. Wealth changes are expected to generate changes in aggregate consumption. These in turn affect firms' sales revenues and consequently (given their costs) their earnings. We expect the sensitivities of firms' earnings to changes in aggregate wealth to differ, which provides a second source of measurable variation in systematic earnings risk.

We conjecture that firm's exposures to aggregate supply shocks differ from their exposures to demand shocks. We document that they do differ, and in fact are negatively correlated. Firms' differential exposures to aggregate supply and demand shocks are expected due to heterogeneity of production functions and of products or services. For example, if a firm's production function yields constant returns to scale and its earnings are mainly driven by the elasticity of demand for its output, the firm's earnings are expected to be more sensitive to demand shocks. If a firm's production function depends on innovation to generate increasing returns to scale regardless of demand, its earnings are expected to be more sensitive to supply shocks. A low-cost manufacturer's production function might be highly dependent on aggregate TFP changes, but its output might be a defensive good and insensitive to fluctuation in total wealth. A luxury brand's demand might be very sensitive to changes in wealth, but its production function might be insensitive to changes in productivity.

This conjecture is consistent with observations in existing studies. For example, when aggregate household wealth (our instrument for aggregate demand shocks) declines, consumers may shift demand from luxury to essential goods (Ait-Sahalia, Parker, and Yogo, 2004), reduce consumption of durable goods, or switch to lower priced products (Bijmolt, van Heerde, and Pieters, 2005). Hoch, Kim, Montgomery, and Rossi (1995) show that elderly and low income households are more sensitive to product prices. Firms have heterogeneous exposures to consumption shifts between luxury and essential goods, between durable and staple goods, or between high and low priced goods. Consequently, the sensitivity of firm-level earnings

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to aggregate demand shocks is expected to differ as a function of firm characteristics such as their clientele profile and product offerings.

Similarly, changes in TFP (our instrument for aggregate supply shocks) are expected to have different firm-level earnings effects. TFP changes measure variation in the aggregate supply of goods and services that cannot be explained by variation in aggregate capital and labor inputs (Comin, 2010), due to productivity-enhancing innovation (Hulten, Dean, and Harper 2007). Papanikolaou (2011) provides evidence that firms' sensitivities to innovation shocks depend on their production functions. For example, a positive innovation shock benefits firms producing investment goods more than firms producing consumption goods. New methods and new technologies are likely to harm or benefit some firms more than others, depending on their production processes. While these studies are related to ours, their empirical methods are substantially different and it is difficult to draw inferences from them for our study. For example, Papanikolaou (2011) examines investment shocks, or innovations to aggregate capital input, using the relative price of new equipment as a proxy. We study TFP shocks, which measure changes in output given the level of investment in capital stocks. TFP controls for the investment shocks studied by Papanikolaou (2011), so in that sense we are studying different phenomena. Viewed in terms of the above Cobb-Douglas production function, we focus on TFP, the A term, while Papanikolaou focuses on K, capital input.

Our thesis that various priced "empirical" factors are associated with firms' differential sensitivities to aggregate demand and supply shocks has loose parallels in production-based and consumption-based asset pricing models. In relation to aggregate supply, Cochrane's (1991, 1996) production and investment based pricing models spawned a literature that includes Zhang (2005), who develops an investment-based asset pricing model linking the book-to-market effect to growth opportunities. Berk, Green, and Naik (1999) argue that a firm's systematic risk is driven by its assets in place and its growth opportunities, which are related to aggregate supply shocks. Carlson, Fisher, and Giammarino (2004) posit that a firm's systematic risk depends on operating leverage, reversible real options, fixed adjustment costs, and finite growth opportunities. They find that the book-to-market effect relates to

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operating leverage and the size effect relates to the residual importance of growth options relative to assets in place. Kogan and Papanikolaou (2014) and Kung and Schmid (2015) both explore how technological innovation affects growth opportunities and relates to the cross-section of stock returns, though they interpret technological innovation differently. Kogan and Papanikolaou (2014) infer innovation from the decline in the quality-adjusted price of investment goods, relative to consumption goods, while Kung and Schmid (2015) infer it from accumulated aggregate R&D expenditure.

Researchers were unable to show that consumption growth explains asset prices, as predicted by the consumption-based CAPM, until Lettau and Ludvigson (2001) used the consumption-wealth ratio as the explanatory variable. Lettau and Ludvigson (2001) argue that the price of consumption risk varies over time, and that conditional consumption growth deflated by wealth explains asset prices while unconditional growth does not. However, Roussanov (2014) shows that although the conditional covariances of high book-to-market portfolios are time-varying as Lettau and Ludvigson (2001) argue, the conditional expected returns of high book-to-market portfolios do not vary with time. He posits that a two-factor conditional consumption-based CAPM explains the book-to-market effect better than the one-factor conditional consumption-based CAPM, the two factors being growth in aggregate consumption and aggregate wealth.

Our study also relates to a stream of literature that examines stock returns and macroeconomic indicators.[6] Imrohoroglu and Tuzel (2014) examine firm-level TFP, while we examine the sensitivity of a firm's earnings to aggregate ΔTFP. Firm-level TFP may be due to many factors, such as unique inputs, proprietary information such as trade secrets, or economies of scale. We are unaware of a direct theoretical relationship between the level of firm TFP and the sensitivity of a firm's earnings to aggregate ΔTFP. Lettau, Ludvigson, and Ma (2019) examine the capital share of national income. For example, salaries contribute the majority of median household income, while capital gains contribute the majority of the wealthiest households' incomes. They focus on the share of capital versus labor, while we examine total

[6] A related literature studies stock returns and macroeconomic variables such as industrial production, default risk premium, and growth in income per capita (e.g., Chen, Roll, and Ross, 1986; Fama and French, 1989; and Jagannathan and Wang, 1996).

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household wealth.

Gomes, Kogan, and Yogo (2009) find that non-durable good sectors are more exposed to systematic risk than durable good and service sectors. In Section 7.6, we use their industry classifications to examine whether these sectors exhibit differential earnings-based systematic risks.

Our approach differs in several ways from the exact decomposition of returns into cash flow shocks and discount rate shocks by Campbell and Vuolteenaho (2004). First, we estimate systematic risk from accounting earnings, not stock returns. Second, our instruments for aggregate demand and supply shocks are not intended to be exhaustive of all aggregate risk; they merely are instruments for it. Nor are aggregate demand and supply shocks intended to be mutually exclusive.

5 Sample and Data

Our sample comprises US firms with listed common equity securities (CRSP share code: 10 or 11), excluding ADRs (American Depository Receipts), REITs (Real Estate Investment Trusts), and closed-end funds. The sample period starts in 1964 to avoid selection bias in the pre-1963 Compustat annual data (Ball and Watts, 1977; Fama and French, 1992) and we need one prior year of accounting information to form portfolios. The sample period ends in 2018.

We initially define annual earnings as operating income (Compustat: OIADP) minus preferred stock cash dividends (PDVC). If operating income is missing, we replace it with net income (Compustat: IB). Aggregate earnings changes is calculated as: $\Delta Earnings_{MKT,t} = [Earnings_{MKT,t} - Earnings_{MKT,t-1}] / P_{MKT,t-2}$ where $Earnings_{MKT,t}$ is the sum of earnings during year t across all sample firms and $P_{MKT,t-2}$ is the sum of their market capitalizations at the end of year $t-2$. [7] Firm-level earnings are winsorized at $+/-1\%$. In Section 7.4, we discuss results based on three alternate earnings measures: operating income after

[7] We use market capitalization in the end of year $t-2$, rather than $t-1$, as a deflator to avoid a mechanical relation with return momentum when we sort firms into portfolios based on stock returns in year $t-1$.

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interest expense, operating income after interest and tax expenses, and operating income before R&D expense (to circumvent the immediate expensing rules of U.S. GAAP).

The instrument for aggregate demand shocks is changes in aggregate household wealth, excluding stocks. Data are from the household net worth series of the Board of Governors of the Federal Reserve (Z1/Z1/FL152090005). In that series, net worth is estimated as the sum of non-financial assets such as real estate and financial assets such as deposits and securities, minus household liabilities such as mortgages. We use aggregate rather than median household wealth because households have more discretion over consumption of luxury goods than essential goods (Ait-Sahalia, Parker, and Yogo, 2004). We exclude changes in equity security values (corporate equities Z1/Z1/FL153064105 and indirectly held corporate equities Z1/Z1/FL153064175), for two reasons. One reason for excluding stocks stems from our interest in the risk information contained in accounting earnings. Equity returns reflect revisions in forward-looking information, and thus are misaligned in time with (i.e., lead) realized earnings and macroeconomic outcomes. Consistent with misalignment, aggregate returns are not significantly associated with contemporaneous aggregate earnings, but rather predict aggregate earnings and real GDP growth in the two following years [Chen (1991),Sadka and Sadka (2009), and Ball, Sadka, and Sadka (2009)]. This time-misalignment would obscure the contemporaneous firm-aggregate relation we are estimating. A second reason for excluding equity investments is that we are focused on the risk in earnings rather than stock returns. The channel through which firms' accounting earnings reflect wealth is through revenues earned from the contemporaneous consumption of their goods and services by households. Households are not expected sell equity investments (including 401Ks) strictly for consumption purposes as a function of short term returns. While equities represent only 15% of total household wealth on average over 1945-2013, their values are so volatile that they dominate other wealth components.

In sum, we expect short term changes in equity values do not drive current consumption and current earnings, and add substantial unrelated volatility to the measure of wealth. Consistent with that expectation, if stocks are included we find that changes in aggregate wealth are insignificantly associated with contemporaneous aggregate earnings and with

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changes in TFP, but significantly associated with their values in the following year. In addition, the earnings betas then are more weakly associated with firm characteristics and future stock returns.

The instrument for aggregate supply shocks is changes in TFP $\Delta \ln TFP = \Delta \ln Y - \alpha \Delta \ln K - (1-\alpha)\Delta \ln L$ (i.e., changes in output ΔY net of changes in the contributions of capital K and labor L), following Fernald (2012).[8]

Aggregate changes in earnings, aggregate ΔWealth , and aggregate ΔTFP are measured with annual frequency, to avoid the noise and seasonality in higher-frequency data (Kothari, Shanken, and Sloan, 1995). In addition, serial correlation is a lesser concern in the annual time series. For example, the serial correlations over the first three lags in ΔTFP are 0.11, -0.09, and -0.19, and in annual aggregate earnings changes they are 0.14, -0.32, and -0.23 (see Table 1 Panel B). However, the serial correlation in annual changes in wealth are 0.74, 0.45, and 0.22, so we use first differences and they fall to 0.11, -0.13, and -0.22 (Table 1 Panel B). The moderate serial correlation in each of the three time series suggests they likely capture aggregate shocks.

The mean, median and standard deviation of the aggregate earnings changes series are 0.025, 0.025, and 0.020 respectively. The equivalent statistics for ΔTFP are 0.009, 0.010, and 0.016; for the adjusted ΔWealth they are 0.000, 0.002, and 0.027. All variables therefore exhibit substantial variation with limited skew (see Table 1 Panel A). In Table 1 Panel C, the correlation between earnings changes and ΔTFP/ΔWealth ranges from 0.25 to 0.44 and the correlation between ΔTFP and ΔWealth is about 0.50, implying that aggregate demand and supply shocks are not independent.

[8]Both the macroeconomic indicators and annual earnings are nominal values, avoiding the potential price-level mismatch discussed in Konchitchki (2011).

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5.1 Aggregate Time-Series Regressions

We first check that aggregate earnings changes is affected by changes in aggregate wealth and aggregate productivity, using contemporaneous time-series regressions of the form:

$$\Delta Earnings_{MKT,t} = \alpha_0 + \alpha_1 \Delta Z_t + \varepsilon_{\alpha,t} \text{ where } Z = TFP, Wealth, \text{or both} \quad (1)$$

where $\Delta Earnings_{MKT,t}$ is annual aggregate earnings change in year t and ΔZ_t is annual change in wealth or TFP in year t. Aggregate earnings are expected to increase in aggregate firm revenue, and hence to increase in aggregate demand. Similarly, aggregate earnings are expected to decrease in aggregate firm costs, and hence to increase in TFP. Therefore, we expect the α_1 estimated from Equation (1) to be positive.

In Table 2 Panel A, we do find that aggregate earnings changes are positively related to both ΔWealth and ΔTFP. The effect of ΔTFP on aggregate earnings changes ($t = 2.88$) is slightly stronger than the effect of ΔWealth ($t = 1.91$). When including both ΔTFP and ΔWealth in the regression, both coefficients remain positive but with reduced statistical significance, potentially due to the correlation between TFP and wealth shocks. Overall, these findings suggest that aggregate earnings changes reflects both aggregate wealth and aggregate productivity shocks, consistent with both macroeconomic shocks contributing to aggregate risk.

We conduct a lead-lag analysis to validate our premise that macroeconomic variables are better aligned in time with earnings than with returns. Results in Table 2 Panel B confirm prior evidence[9] that aggregate returns are not significantly associated with contemporaneous aggregate earnings but predict earnings in the two following years. Results in Table 2 Panels C-F show that aggregate returns also convey forward-looking information about changes in GDP, aggregate consumption, TFP, and aggregate wealth. In contrast, results in Table 2 Panels G-H show that aggregate earnings are associated only with contemporaneous values of ΔTFP and ΔWealth. Collectively, these results validate our premise that

[9] For example, Chen (1991), Sadka and Sadka (2009) and Ball, Sadka, and Sadka (2009).

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macroeconomic variables are better aligned in time with earnings than with returns. Given the choice of macroeconomic variables rather than public-firm indexes to measure aggregate risk, their better alignment with earnings suggests their pairing offers better identification of the comovements that underlie systematic risk.[10]

We further conduct a principle component analysis to examine the incremental information ΔTFP and ΔWealth capture beyond aggregate "index" earnings. Following Chen, Roll, and Ross (1986), we select eight macroeconomic outcomes: 30-day Treasury-bill rate, term spread (30-year Treasury-bond rate minus 30-day Treasury-bill rate), default spread (Baa corporate bond yield minus 30-year Treasury-bond rate), real GDP growth, change in the unemployment rate, change in consumption per capita, change in housing starts, and change in the oil price per barrel. Data comes from CRSP, Federal Reserve Bank of St. Louis, and US Energy Information Administration. Annual series is from 1964 to 2018 (N=55). We reduce the eight economic outcome series to four orthogonalized components in which eigenvalues are above one. We find that changes in TFP and wealth are significantly more correlated with the macroeconomic outcome principal components than aggregate earnings, based on publicly listed firms. The finding validates the argument that the macroeconomic indicators we select reflect a substantially wider set of economic outcomes, such as GDP, unemployment, consumer spending, and housing starts, than public-firm indexes, and hence that shocks to the indicators provide better measures of systematic risk.

6 Firm Characteristics and Sensitivity to Demand and Supply Shocks

We select four firm characteristics that have been documented to be affected by macroeconomic shocks: book-to-market, return momentum, size, and asset growth. Fama and French (1989) show that default risk premium and aggregate dividend yield have a greater impact on

[10] Konchitchki and Patatoukas (2014a, b), find that quarterly earnings predict future quarterly GDP and other macroeconomic indicators. In annual data, we do not find that annual ΔEarnings predict future annual ΔGDP, ΔTFP, or ΔWealth.

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future equal-weighted aggregate stock returns than value-weighted stock returns, suggesting that small firms are more sensitive to macroeconomic conditions. Gomes, Kogan, and Zhang (2003) demonstrate analytically that size captures the component of a firm's systematic risk related to its growth options, whereas the book-to-market ratio is a measure of the risk of the firm's assets in place. Petkova and Zhang (2005) find evidence that the book-to-market ratio varies with measures of the expected market risk premium. Chordia and Shivakumar (2006) find that the ability of return momentum to explain cross-sectional stock returns is subsumed by the earnings momentum spread, which is related to future macroeconomic conditions, such as growth in GDP, industrial production, consumption, labor income, inflation and T-bill returns. Cooper, Gulen and Schill (2008) use asset growth to capture firm investment and Hubbard, Kashyap, and Whited (1993) show that firm investment is sensitive to macroeconomic conditions, measured by the default risk premium. Moreover, Papanikolaou (2011) formalizes the relation between firm investment and macroeconomic conditions and argues that firm-level sensitivity to macroeconomic shocks depends on their production functions. These studies suggest that book-to-market, return momentum, size, and asset growth could be good instruments for firm-level sensitivity to macroeconomic shocks. For each characteristic, we sort firms annually into five portfolios based on its most recent value, and regress portfolio-level earnings changes onΔTFP and ΔWealth. We report earnings-based supply and demand betas as a function of portfolio rank.

6.1 Book-to-Market Portfolios and Demand and Supply Betas

Book-to-market predicts average returns (e.g., Rosenberg, Reid, and Lanstein, 1985; Fama and French, 1992; Cohen, Polk, and Vuolteenaho, 2003; Asness, Moskowitz, and Pedersen, 2013). We sort firms into five portfolios based on *B/M,* defined as book equity at the end of year *t-1* divided by market equity (CRSP: PRC, SHROUT). Book equity is total assets (Compustat: AT) minus liabilities (LT), plus balance sheet deferred taxes and investment tax credit (TXDITC), minus preferred stock liquidating value (PSTKL), redemption value (PSTKRV), or carrying value (PSTK). Then we estimate the following time-series regression

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for each portfolio:

$$\Delta Earnings_{t,p} = \beta_{0p} + \beta_{1p}\Delta Z_t + \varepsilon_{\beta,t,p} \text{ where } Z = TFP, Wealth, \text{ or both} \quad (2)$$

Portfolio earnings changes $\Delta Earnings_{t,p}$ is calculated as changes in earnings scaled by market values. Specifically, $\Delta Earnings_{t,p} = [Earnings_{t,p} - Earnings_{t-1,p}] / P_{t-2,p}$ where $Earnings_{t,p}$ is the sum of earnings during year t for all portfolio firms and $P_{t-2,p}$ is the portfolio market capitalization in the end of year t-2. If the book-to-market "value premium" is due to higher systematic risk, we expect high book-to-market firms' earnings to be more sensitive to demand and supply shocks, captured by contemporaneous ΔWealth and ΔTFP.

The results in Table 3 indicate that sensitivities to TFP and wealth shocks are both increasing in B/M. The sensitivity of earnings to productivity shocks monotonically decreases from the high B/M portfolio (1.352) to the low B/M portfolio (0.077).[11] The sensitivity of earnings to wealth shocks also monotonically decreases from the high B/M portfolio (0.550) to the low B/M portfolio (0.031). When estimated jointly, both earnings betas are still increasing in B/M, suggesting that the supply effect does not subsume the demand effect or vice versa even though annual ΔTFP and $\Delta Wealth$ are highly correlated. The adjusted R^2 also monotonically decreases from the high B/M portfolio (0.24) to the low B/M portfolio (-0.02).

These findings are consistent with the thesis that expected returns increase in B/M at least in part because B/M is related to systematic earnings risk (see also Fama and French, 1995; Campbell and Vuolteenaho, 2004; Ball, Sadka, and Sadka, 2009; Campbell, Polk, and Vuolteenaho, 2010; Ellahie, Katz, and Richardson, 2020; Ball, Gerakos, Linnainmaa, and Nikolaev, 2020). To test robustness, we replace $P_{t-2,p}$ with total assets $TA_{t-2,p}$ or with book equities $BVE_{t-2,p}$ as the earnings changes deflator and find similar results. Earnings-based supply and demand betas both increase monotonically in B/M, as does the adjusted R^2. In sum, we find robust evidence that the value premium is associated with systematic risk

[11] This result is consistent with the finding of Andersen, Bollerslev, Diebold, and Wu (2005) that the covariance between portfolio market beta and lagged industrial production growth increases from the low B/M portfolio to the high B/M portfolio.

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arising from aggregate productivity and wealth shocks.

6.2 Return Momentum Portfolios and Demand and Supply Betas

Return momentum is another firm characteristic associated with future stock returns (e.g., Jagadeesh and Titman, 1993; Asness, Moskowitz, and Pedersen, 2013). We employ the same procedure as for B/M to examine whether this characteristic is associated with sensitivity to aggregate demand and supply shocks. We sort firms into five portfolios based on cumulated stock returns from $j-2$ to $j-12$, where j is the portfolio formation month.

In Table 4, we document that earnings-based supply and demand betas are increasing in return momentum, although not always monotonically. The sensitivity of earnings to wealth shocks decreases from the high momentum portfolio (0.484) to the low momentum portfolio (0.242), and the sensitivity of earnings to productivity shocks decreases from the high momentum portfolio (0.883) to the low momentum portfolio (0.444). When including both demand and supply shocks in the regression, we continue to find that sensitivities to wealth and productivity shocks are increasing in return momentum. These results suggest that return momentum may be related to exposure to systematic risks. However, we note that the adjusted R^2 values are a U-shaped function of portfolio momentum rank. We further caution that our results may be unique to the deflator, market price at the end of year $t-2$, because sensitivity to wealth shocks is no longer increasing in return momentum when portfolio earnings are deflated by total assets or book equities.[12]

6.3 Size Portfolios and Demand and Supply Betas

Since Banz (1981) documented a size effect in stock returns, size has been a staple in empirical asset pricing models. For example, it is one of three factors in the Fama and French (1995)

[12]The result of earnings wealth betas increasing in return momentum is potentially driven by recoveries from economic downturns. Firms' market capitalizations are low in downturns and earnings growth increases in recoveries. Deflating earnings change $[Earning_{t,p} - Earning_{t-1,p}]$ by lagged market value $P_{t-2,p}$ thus could amplify high return momentum firms' sensitivities to wealth shocks. The result is not observed when deflating by book assets or book equity, which are less depressed in downturns, or by smoothed market capitalization (e.g., a three-year or five-year average).

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model. Small firms yield higher average returns than large firms. We explore whether this firm characteristic is associated with sensitivity to aggregate demand and supply shocks.

We sort firms into five portfolios based on market capitalization (CRSP: PRC, SHROUT) at the end of year *t-1* and estimate supply and demand betas from time-series regressions for each portfolio as in Equation (2). In Table 5, we find that earnings sensitivities to productivity and wealth shocks are both decreasing in firm size. Earnings-based sensitivity to productivity shocks (supply betas) increases monotonically from the large firm portfolio (0.294) to the small firm portfolio (1.267). Earnings-based sensitivity to wealth shocks (demand betas) increases monotonically from the large firm portfolio (0.128) to the small firm portfolio (0.585). When estimated in a single regression, both demand and supply betas still decrease in firm size, suggesting that firm size captures systematic risks regarding demand and supply shocks.

In robustness checks, earnings sensitivity to productivity shocks still decreases in firm size when earnings changes is deflated by total assets $TA_{t-2,p}$ or book equities $BVE_{t-2,p}$ instead of $P_{t-2,p}$, but the relation between wealth betas and size is weaker, again consistent with the Table 5 result that the impact of supply dominates that of demand. In sum, these findings suggest that the higher returns earned by small firms may be due to their higher sensitivity to productivity shocks.

6.4 Asset Growth Portfolios and Demand and Supply Betas

Cooper, Gulen, and Schill (2008) present evidence that higher asset growth firms earn lower average returns. To examine whether this anomaly is associated with exposure to aggregate demand and supply shocks, we sort firms into five portfolios based on change in the natural log of book value of assets per split-adjusted share in the prior year.

Results from the time-series regressions based on Equation (2) indicate that sensitivities to productivity and wealth shocks are both decreasing in asset growth. In Table 6, the sensitivity of earnings changes to productivity shocks increases monotonically from the high asset growth portfolio (0.227) to the low asset growth portfolio (0.813). The sensitivity

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of earnings changes to wealth shocks also increases monotonically from the high growth portfolio (0.072) to the low growth portfolio (0.423). When including both demand and supply shocks in the regression, both earnings betas still decrease in asset growth.

In robustness tests, earnings-based supply and demand betas continue to monotonically decrease in asset growth when earnings changes is deflated by total assets $TA_{t-2,p}$ or book equities $BVE_{t-2,p}$ instead of $P_{t-2,p}$. In sum, we find consistent evidence that asset growth is associated with a reduction in firm exposure to aggregate demand and supply shocks.

6.5 Summary

We find that systematic risk—in the form of firm-level exposure to aggregate demand and supply shocks—is associated with four firm characteristics that are known to predict stock returns: book-to-market, return momentum, size, and asset growth. These results could provide some insight into why the firm characteristics are priced, a conjecture we test below. We caution that we fail to find similar results for portfolios formed on earnings-to-price (e.g., Basu, 1977), post-earnings-announcement drift (e.g., Ball and Brown, 1968; Bernard and Thomas, 1989, 1990), profit margin (Novy-Marx, 2013; Ball, Gerakos, Linnainmaa, and Nikolaev, 2015, 2016), or retained earnings to market equity (Ball, Gerakos, Linnainmaa, and Nikolaev, 2020).

7 Pricing Tests

In this section, we estimate earnings-based supply betas (sensitivity to aggregate TFP shocks) and demand betas (sensitivity to aggregate wealth shocks), and show that they are negatively correlated. Both TFP and wealth betas generate positive risk premia, in the sense that they help explain the cross-section of future stock returns. Finally, we examine the extent to which the explanatory power of firm characteristics could be due to their correlation with our earnings-based supply and demand betas. We caution that firm-level monthly returns exhibit substantial volatility, and hence cross-sectional R^2s are small. Following

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convention, we report t-statistics estimated using the Fama and MacBeth (1973) procedure. We also report test statistics based on Newey-West (1994) standard errors, as well as GLS regressions as suggested by Lewellen, Nagel, and Shanken (2010).

7.1 Earnings Beta Estimates

Because the goal is to identify systematic risk using annual earnings, we are constrained by their low frequency. We choose annual earnings to avoid the noise and seasonality in quarterly data (Kothari, Shanken, and Sloan, 1995). Unlike high-frequency stock returns (e.g., monthly, daily, or intraday), we do not have enough earnings observations to reliably estimate firm-level earnings betas. For example, if we require a minimum of only 20 annual earnings observations, we impose a substantial survivorship bias. Even then, an earnings beta estimate based on only 20 observations confronts the fact that earnings are left-skewed. The skew is in large part due to accounting rules that require companies to write down assets when their market values are lower than their historical costs, but that do not permit writing up assets when their market values exceed historical costs. Due to these unique earnings properties, we choose to estimate earnings betas at the portfolio level and then assign betas to individual firms based on their portfolio membership.

We start the beta estimation process by assigning firms to 81 portfolios, based on the terciles of each firm characteristic. We choose terciles rather than quintiles as a trade-off between the number of portfolios and the number of firms in each portfolio. A larger number of portfolios provides greater cross-sectional variation in beta estimates but induces greater estimation error due to the small number of firms in each portfolio. There are, on average, 5,700 firms in a year. We choose terciles for each firm characteristic and create a total of 81 portfolios, so on average, there are about 70 firms in a portfolio.

Next, we use all 55 years from 1964-2018 to run two time-series regressions and obtain 81 pairs of index, TFP, and wealth betas. In the first time-series regression, we estimate an earnings index beta for each portfolio as follows:

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$$\Delta Earnings_{t,p} = \gamma_{0p} + \beta_{MKT,p}\Delta Earnings_{MKT,t} + \varepsilon_{\gamma,t,p} \tag{3}$$

Portfolio earnings changes $\Delta Earnings_{t,p}$ is $[Earnings_{t,p} - Earnings_{t-1,p}]/P_{t-2,p}$ where $Earnings_{t,p}$ is the sum of year t earnings of all firms in portfolio p and $P_{t-2,p}$ is total portfolio market capitalization in the end of year $t-2$. $\Delta Earnings_{MKT,t}$ is annual aggregate earnings changes of all sample firms for calendar year t, calculated similarly. $\beta_{MKT,p}$ is the earnings index beta of portfolio p.

In the second time-series regression, we estimate earnings-based wealth and productivity betas using the following bivariate regression for each portfolio:

$$\Delta Earnings_{t,p} = \delta_{0p} + \beta_{Wealth,p}\Delta Wealth_t + \beta_{TFP,p}\Delta TFP_t + \varepsilon_{\delta,t,p} \tag{4}$$

$\beta_{Wealth,p}$ is the earnings wealth beta of portfolio p (sensitivity to aggregate wealth shocks) and $\beta_{TFP,p}$ is the portfolio's earnings productivity beta (sensitivity to aggregate productivity shocks). ΔWealth and ΔTFP are included in a single regression because they are positively correlated, approximately 0.5, implying that the variables share approximately one quarter of their variation (see Table 1 Panel C). Nevertheless, there remains a substantial amount of their variation that is not shared, consistent with them being somewhat different economic factors. Because they are correlated, we regress changes in earnings on both ΔTFP and ΔWealth to estimate $\widehat{\beta}_{TFP}$ and $\widehat{\beta}_{Wealth}$ jointly, and we acknowledge that it is not possible to fully disentangle their relationships with earnings.

Table 7 Panel A summarizes the distributions across the 81 portfolios of the estimated earnings-based index, supply and demand betas.[13] The three distributions are right skewed. The large arithmetic mean of annual index beta is due largely to small and low asset growth firms. Panel B indicates that index beta is positively correlated with both wealth beta and TFP beta ($\rho = 0.35$ and 0.43 Spearman respectively), consistent with both macroeconomic

[13]Unlike a "market model" regression in which the independent variable (the "market" index) is the mean of the dependent variables, and hence the mean beta is unity, the means of our two systematic risk instruments are not constrained.

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shocks contributing to systematic risk at the firm level. An important result is that TFP beta and wealth beta are negatively correlated (ρ = -0.46 Pearson or ρ = -0.23 Spearman), implying that firms that are more (less) sensitive to aggregate demand shocks are less (more) sensitive to aggregate supply shocks.

We further conduct an industry analysis to validate the notion that systematic earnings risk originates in different economic forces across firms. Specifically, we use the MSCI's Global Industry Classification Standard six-digit and eight-digit industries to form portfolios and estimate three earnings betas: Index beta, TFP beta, and wealth beta. We continue to find a negative correlation between TFP and wealth betas, suggesting that industries with a high TFP beta have a low wealth beta, and vice versa. For example, copper, electronic components, software, airports, marine ports, highways, and railroad industries have a high TFP beta and a low wealth beta, while thrifts and mortgage finance, leisure facilities, and consumable fuel industries have a high wealth beta and a low TFP beta. The negative correlation between TFP and wealth betas is more pronounced when using 186 GICS eight-digit industry portfolios than 76 GICS six-digit industry portfolios. The industry analysis suggests the usefulness of two earnings betas, based on TFP and wealth.

Finally, we assign the three earnings beta estimates ($\widehat{\beta}_{MKT,p}, \widehat{\beta}_{Wealth,p}, \widehat{\beta}_{TFP,p}$) to individual firms based on their annual portfolio membership. Although portfolio-level betas are constant, firm-level betas vary over time because firm change portfolios (and hence betas) as frequently as annually, due to changes in their four firm characteristics. We find that 10.31% of the sample firms remain in the same portfolio for two years and 1.66% of firms stay in the same portfolio for three years. Moreover, the serial correlation of firm-year earnings betas presented in Table 7 Panel C suggests that our estimated earnings betas vary across firm-years.

7.2 Asset Prices and Earnings Betas

We estimate the relation between earnings betas and monthly returns using the following procedure. Returns are measured with a six-month lag to avoid any market reaction to the

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prior year-end values of the four firm characteristics. For example, for portfolios formed at the end of 2001, we examine firm-level twelve-month returns from July 2002 to June 2003. In this pricing test, returns vary monthly but betas vary annually.

In each month, we estimate two cross-sectional regressions:

$$RET_{m,i} = \eta_{0m} + \eta_{1m}\widehat{\beta}_{MKT,t,i} + \varepsilon_{\eta,m,i} \text{ and } RET_{m,i} = \lambda_{0m} + \lambda_{1m}\widehat{\beta}_{TFP,t,i} + \lambda_{2m}\widehat{\beta}_{Wealth,t,i} + \varepsilon_{\lambda,m,i} \tag{5}$$

$RET_{m,i}$ is the return in month m for firm i and $\widehat{\beta}_{MKT,t,i}, \widehat{\beta}_{Wealth,t,i}, \widehat{\beta}_{TFP,t,i}$ are earnings betas for firm i in year t.[14] The regression slopes η_{1m}, λ_{1m} and λ_{2m} measure the ex post risk-return relations in month m, and their time-series average values are interpreted as risk premiums.[15]

In Table 8 Panel A we report average monthly intercepts and slope coefficients, along with Fama and MacBeth (1973) t-statistics and average regression adjusted R^2s. In Model (1), the average risk premium for the single earnings-based index-beta factor is positive and significant (t-stat = 5.66). As earnings index beta moves from the 25th to the 75th percentile, firm-level monthly return increases on average by 0.28%. These results are consistent with those in Da and Warachka (2009) and Ellahie (2020).

Recall that our thesis is that an index of public-firm earnings provides an inferior measure of aggregate risk because there are abundant other assets into which investors are diversified. As a move in the direction of incorporating the aggregate risk generated by all assets, in Model (2) we replace $\widehat{\beta}_{MKT}$ in the cross-sectional regressions with $\widehat{\beta}_{TFP}$ and $\widehat{\beta}_{Wealth}$. These represent firm-level sensitivities to aggregate economy-wide demand and supply shocks, not simply to the performance of listed equities. The time-series averages of λ_{1m} and λ_{2m} (the cross-sectional slopes for TFP and wealth betas) are both positive and significant (Fama and MacBeth t-stats = 5.47 and 7.30 respectively). As TFP beta and wealth beta move

[14] More precisely, $\widehat{\beta}_{MKT,t,i}, \widehat{\beta}_{Wealth,t,i}, \widehat{\beta}_{TFP,t,i}$ are earnings betas for firm i in year $t-1$ because we use firm characteristics in year $t-1$ to assign earnings betas to individual firms. We then examine their relation with the twelve monthly returns from July of year t to June of year $t+1$. We use year t in the subscript in the equation for brevity.

[15] The "risk premiums" for TFP and wealth betas are not the traditional tradable premiums because both TFP and wealth betas are unlikely to be observable in real time. Therefore, we are not able to construct factor mimicking portfolios. Our objective is not to document a new trading factor, but rather to understand the nature of the risk information contained in accounting earnings.

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from their 25th to 75th percentiles, firm-level monthly return increases on average by 0.29% and 0.18% respectively, indicating that these premia are economically as well as statistically significant. The adjusted R^2 approximately doubles, from 0.0032 in the single-beta model to 0.0063 in the two-variable model. The two-beta model's adjusted R^2 is significantly greater than the index beta model's adjusted R^2 (p-value<0.01). Due to the high volatility of monthly individual-firm returns, the explanatory power of cross-sectional pricing regressions invariably is low.

To address potential serial correlation among the monthly error terms when earnings betas are updated only annually, we estimate t-statistics based on Newey-West (1994) standard errors adjusting for twelve monthly lags. Earnings index, TFP, and wealth betas continue to present statistically significant risk premia, although the magnitudes of the t-statistics are smaller (Newey-West t-stats $t_{NW} = 4.59$, 4.66, and 5.99 respectively).

To address the concern that test statistics are inflated because rebalancing portfolios annually causes betas to repeat for twelve monthly regressions, we conduct a pricing test using annual returns. Specifically, a firm's characteristics at the end of year t-1 determine its assignment to a portfolio and hence its beta estimates. Twelve subsequent monthly returns are merged with these betas, starting in the seventh month after portfolio formation. Then we calculate the annual return for each firm and run the cross-sectional pricing tests for each year. The annual test t-statistics are lower than the monthly test t-statistics (annual t-stats $t_{Ann} = 2.24$, 4.26, and 1.90 for index, TFP, and wealth betas respectively). The considerably weaker t-statistics for wealth beta suggests that demand shocks are potentially shorter-lived than supply shocks. The annual test adjusted R^2_{Ann} (1.03%) is higher than the monthly test adjusted R^2 (0.63%), suggesting that monthly returns are noisier than annual returns. We also find that the annual test adjusted R^2 from the bivariate TFP and wealth betas model almost doubles, compared to the annual test adjusted R^2 from the single index beta model (0.53%). The two-beta model's adjusted R^2 is significantly greater than the index beta model's adjusted R^2 (p-value<0.01).

Lastly, we follow the suggestion by Lewellen, Nagel, and Shanken (2010) and report GLS R^2s (3.60% for the single index beta model and 2.89% for the bivariate TFP and wealth betas

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model). The higher GLS R^2s also suggest firm-level monthly returns are noisy. Moreover, we no longer observe an improvement in the explanatory power by the bivariate TFP and wealth betas model. Nevertheless, the significant coefficients on both supply and demand betas suggest that the two types of firm systematic risks explain the cross-section of stock returns.[16]

7.3 Asset Prices, Firm Characteristics, and Earnings Betas

An interesting question is how much of the ability of the four firm characteristics we study (book-to-market, return momentum, size, and asset growth) to predict returns originates in their correlation with underlying economic sources of systematic risk, and specifically in their correlation with sensitivity to TFP and wealth shocks. We first run a cross-sectional regression of future monthly returns on lagged firm characteristics to provide a benchmark adjusted R^2:

$$RET_{m,i} = \theta_{0m} + \theta_{1m} log(B/M)_{t,i} + \theta_{2m} MMT_{t,i} + \theta_{3m} log(SIZE)_{t,i} + \theta_{4m} ATG_{t,i} + \varepsilon_{\theta,m,i} \quad (6)$$

$RET_{m,i}$ is the return in month m for firm i and $log(B/M)_{t,i}$, $MMT_{t,i}$, $log(SIZE)_{t,i}$, and $ATG_{t,i}$ are firm i characteristics in the end of year $t-1$ prior to month m. In Table 8 Panel B, we report the average monthly intercept and slope coefficient, Fama and MacBeth (1973) t-statistics and the average regression adjusted R^2, and test statistics based on the Newey-West (1994) standard errors, annual return regressions, and GLS regressions. We designate this as Model (3). Consistent with prior evidence, B/M carries a positive risk premium and the other three characteristics carry a negative risk premium.[17] The model explains 2.42% of

[16] We thank an anonymous reviewer for these helpful suggestions.

[17] Because we use twelve-month cumulative returns from December of year t-2 to November of year t-1 to identify return momentum and then link with monthly returns from June of year t+1 to July of t+2, cumulative returns start presenting long-run reversal (i.e., negative risk premiums). This is consistent with De Bondt and Thaler (1985), (1987) and Chopra, Lakonishok, and Ritter (1992) documenting long-term reversal over 2- to 5-year horizons and Jegadeesh and Titman (1993), (2001), Rouwenhorst (1998), (1999), and Chui, Titman, and Wei (2000) documenting short-run momentum over 3- to 12-month horizons.

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the cross-sectional variation in returns (7.99% based on a GLS regression and 3.17% based on an annual return regression).

Next, we orthogonalize each firm characteristic into two components: one related to the TFP and wealth betas and the other unrelated. Specifically, in each year, we regress a firm characteristic on the two earnings betas:

$$K_{t,i} = \omega_{k,0t} + \omega_{k,1t}\widehat{\beta}_{TFP,t,i} + \omega_{k,2t}\widehat{\beta}_{Wealth,t,i} + \varepsilon_{\omega k,t,i} \tag{7}$$

Here $K_{t,i}$ represents firm characteristic K for firm i in year t. In each year, we run a cross-sectional regression and use the estimated $\widehat{\omega}_{k,0t}$, $\widehat{\omega}_{k,1t}$, and $\widehat{\omega}_{k,2t}$ to calculate the residual $\widehat{\varepsilon}_{\omega k,t,i}$ for each firm characteristic k. Each firm then has four estimated residuals in a given year: $\widehat{\varepsilon}_{log(B/M)}$, $\widehat{\varepsilon}_{MMT}$, $\widehat{\varepsilon}_{log(SIZE)}$, and $\widehat{\varepsilon}_{ATG}$.These represent the components of the four firm characteristics that are unrelated to the TFP and wealth betas.

Then we regress the same set of firm-level monthly returns on the four estimated residuals:

$$RET_{m,i} = \nu_{0m} + \nu_{1m}\,\widehat{\varepsilon}_{log(B/M),t,i} + \nu_{2m}\,\widehat{\varepsilon}_{MMT,t,i} + \nu_{3m}\,\widehat{\varepsilon}_{log(SIZE),t,i} + \nu_{4m}\,\widehat{\varepsilon}_{ATG,t,i} + \varepsilon_{\nu,m,i} \tag{8}$$

This specification avoids the concern raised in Chen, Hribar, and Melessa (2018) because the first-step residuals are independent variables, rather than dependent variables, in the second-step regression. Moreover, because Model (6) and Model (8) share the same dependent variable, the adjusted R^2s of the two models are comparable.

We find that t-statistics for the four firm characteristics generally decrease. Collectively, approximately a quarter of the explanatory power is reduced. The Fama-MacBeth R^2 falls from 2.42% to 1.83%, the GLS R^2 falls from 7.99% to 5.74%, and the annual return test R^2 falls from 3.71% to 2.76%. This finding provides an important clue as to why "empirical" (i.e., not derived from asset pricing theory) firm characteristics are priced: their ability to predict returns appears to be due at least in part to their correlation with underlying economic sources of systematic risk.

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7.4 Alternate Earnings Measures

In this section, we investigate the sensitivity of the results to varying the definition of operating income, a variable that is net of depreciation and amortization but excludes many components of "bottom line" net income. Those components tend to be transitory or discretionary in nature, and add noise to the earnings variable (Novy-Marx, 2013; Ball, Gerakos, Linnainmaa, and Nikolaev, 2015, 2020).

Operating income is calculated without deducting interest expense, and hence is a levered earnings variable. To investigate the effect of leverage, we study operating income after deducting interest expense and operating income after deducting both interest and income tax expense (interest is deductible from taxable income). Another concern is that the results for TFP betas could be attenuated by the effect of accounting rules that require immediate expensing against operating income of internally-incurred R&D expenditures. If internal R&D expenditures are increasing in shocks to TFP, and are immediately deducted from earnings, the accounting treatment would attenuate the $\widehat{\beta}_{TFP}$ regression slope.[18] To investigate this possibility, we also study operating income after depreciation but before deducting R&D expense.

In aggregate time-series regressions similar to those in Table 2, Panel A, we find that the unlevered earnings both exhibit higher time-series correlation with aggregate ΔTFP and ΔWealth. Nevertheless, the cross-portfolio earnings sensitivities to ΔTFP or Δwealth and the cross-sectional pricing tests yield similar results regardless of which of the three earnings measures is employed. Specifically, we continue to find that high book-to-market firms' earnings are more sensitive to ΔTFP and ΔWealth than low book-to-market firms, high return momentum firms' earnings are more sensitive to ΔTFP and ΔWealth than low return momentum firms, small firms' earnings are more sensitive to ΔTFP and ΔWealth than large firms, and low asset growth firms' earnings are more sensitive to ΔTFP and ΔWealth than high asset growth firms. Although there is a scale effect on the estimated earnings betas, the standard deviations of the three alternate earnings measures are similar.

[18] We thank Stephen Penman for bringing this possibility to our attention.

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More importantly, all three measures yield a negative correlation between TFP beta and wealth beta, and both betas continue to carry a positive risk premium.

In sum, the adjustments for financial leverage, corporate income tax, and R&D expenditure create scale effects but do not alter the basic results.

7.5 Effect of Firm-level TFP

A closely related study is İmrohoroğlu and Tüzel (2014), who study firm-level TFP, as distinct from firm-level exposure to aggregate TFP shocks. They find that firm-level TFP is negatively correlated with book-to-market, positively correlated with size, and (in a Fama and MacBeth procedure) negatively correlated with future returns. The signs of these results for firm-level TFP reverse those we obtain for TFP beta.[19]

İmrohoroğlu and Tüzel (2014, p.2073) argue that "Firms with low productivity are more vulnerable to business cycles and end up being riskier than high productivity firms." Consistent with this expectation, we find that TFP and TFP betas are negatively correlated. In pooled data, the Pearson correlation is -0.25 (p<0.01) and the Spearman is -0.35 (p<0.01). The Pearson time-series correlation between the annual cross-sectional means of the variables is 0.31 (p<0.01) and the Spearman is 0.43 (p<0.01). Using a Fama MacBeth procedure, the correlation is -0.40 (p<0.01). The question that arises is whether the level of TFP in their study is proxying for supply-side systematic risk (measured by TFP beta), or vice versa.

A suspicion that systematic risk is involved in their results is raised by the fact that İmrohoroğlu and Tüzel (2014) report that the relation between the level of TFP and future returns is cyclical, with the premium on low-TFP firms widening on upturns. This cyclically suggests that systematic risk is involved in their results, and in the negative correlation of TFP with TFP beta.

[19]These results are for specifications in which TFP is lagged (i.e., observed prior to stock returns). İmrohoroğlu and Tüzel (2014) find that TFP is positively correlated with contemporaneous returns, which is hardly surprising. TFP here is firm-level relation between inputs and outputs, and structurally related to earnings. Specifically, they measure TFP based on operating income before depreciation and labor expenses, and then adjust it for the number of employees and the net book value of property, plant, and equipment. It is well known that earnings are correlated with contemporaneous returns (Ball and Brown, 1968).

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To investigate the roles of TFP level and TFP beta in predicting returns, in Table 9 we report results from Fama and MacBeth regressions incorporating TFP in the regressions reported in Table 8 Panel A. The sample is firms with Compustat data required to calculate TFP (excluding financials and utilities, and ending in 2013).[20] Results are for monthly returns; there is no substantive difference obtained from an annual return specification. By itself, TFP has the expected negative sign, but is not significant (t = -1.43). When TFP is added to the single (TFP) beta regression, its t-value is -0.04. When added to the two-beta regression, its t-value is 0.96. We interpret these results as indicating that the level of TFP is associated with future returns because it is correlated with TFP betas, that is with sensitivity to aggregate TFP shocks.

7.6 Gomes et al. (2009) Industry Classifications

Our conjecture that whether individual firms are affected more by aggregate supply or aggregate demand shocks depends on the characteristics of production functions and output products/services is closely related to the theory in Gomes, Kogan, and Yogo (2009). They focus on the consumer sector and argue that firms producing durable goods are riskier than those offering services or non-durable goods.

We use their industry classifications to sort sample firms into five groups, based on whether they produce durable consumer goods, non-durable consumer goods, consumer services, investment or intermediate goods (e.g., semiconductor chips or manufacturing equipment), or whether their output is related to public bodies (e.g., infrastructure, roads, airports). Retail, wholesale, and financial firms (SIC 5xxx and 6xxx) are excluded from the Gomes et al. classifications.

In untabulated results, we find that the government sector has the highest earnings sensitivity to ΔTFP (with p-values for differences from all other sectors less than 0.05),

[20] Requiring a positive number of employees (Compustat: EMP) loses 27 percent of the observations because many firms do not disclose that information. Requiring positive values for sales revenue, total assets, capital expenditure, gross PP&E, and accumulated depreciation (Compustat: REVT, AT, CAPX, PPEGT, DPACT) loses a further 11 percent of the observations.

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consistent with the long duration of public infrastructure construction projects and with a lower ability to react quickly to change due to bureaucratic involvement. For ΔWealth, the earnings of all three consumer sectors (durable, non-durable, and service) are more sensitive to changes in wealth than the earnings of the investment good sector (with a p-value for the difference ranges from 0.06 to 0.14). This exercise is consistent with our conjecture that the characteristics of a firm's outputs shape how sensitive the firm's earnings are to aggregate demand and supply shocks.

7.7 Conditional Analysis of Productivity

Motivated by Zhang (2005) and Balvers and Huang (2007), we conduct a conditional analysis of productivity to examine whether the explanatory power of ΔTFP and ΔWealth earnings betas differs between high and low productivity states. We sort 55 years of our sample period from 1964 to 2018 alternatively into a dichotomy, terciles, quartiles, or quintiles, based on ΔInTFP or ΔWealth. Change in total factor productivity ΔlnTFP is ΔlnY-αΔlnK-(1-α)ΔlnL estimated by Fernald (2012). ΔWealth is real GDP growth, adjusted based on an AR(1) process. We run the cross-sectional pricing test for the high and low states separately. In each month, we estimate the following cross-sectional regression: $RET_{m,i} = \lambda_{0m} + \lambda_{1m}\widehat{\beta}_{TFP,t,i} + \lambda_{2m}\widehat{\beta}_{Wealth,t,i} + \varepsilon_{\lambda,m,i}$.

In untabulated results, the two earnings betas model yields a significantly larger adjusted R^2 in a high productivity state than in a low productivity state (p-value < 0.01).[21] For example, the adjusted R^2s for the highest ΔInTFP dichotomy, tercile, quartile, and quintile states are 0.72%, 0.89%, 0.98%, and 1.07%, respectively, compared to a range of 0.53% to 0.59% for the lowest ΔInTFP states. Similarly, the adjusted R^2s for the highest ΔGDP dichotomy, tercile, quartile, and quintile states are 0.67%, 0.81%, 0.92%, and 1.09%, respectively, compared to a range of 0.58% to 0.67% for the lowest ΔGDP states. These results using earnings betas are consistent with the evidence in Balvers and Huang (2007) that value firms are risker in the high productivity risk premium state, thereby providing

[21] We report additional results in our online appendix. It is available at: https://personal.utdallas.edu/~gxs143630/AppendixBST.pdf.

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further validation of earnings betas as measures of systematic risk.

7.8 Out-of-Sample Tests

To address the concern of using the full sample period from 1964-2018 to estimate earnings betas and conduct cross-sectional pricing tests, we conduct two out-of-sample analyses. First, the beta are estimated from 1964-1990 data and the cross-sectional pricing test is conducted over 1991-2018. In untabulated results, we find that TFP and wealth betas are still negatively correlated. Further, we find that both TFP and wealth betas are positively associated with cross-sectional stock returns and the two-beta model's adjusted R^2 is significantly greater than the index beta model's adjusted R^2 (p-value < 0.01). Second, when reversing the procedure with a beta estimation period of 1991-2018 and the pricing test is over 1964-1990, the inferences do not alter.

7.9 Additional Industry Portfolios in Pricing Tests

As suggested by Lewellen, Nagel, and Shanken (2010), one way to improve asset pricing tests is to add industry portfolios to the existing firm characteristic portfolios. For example, they add Fama French 20 industry portfolios to the 25 size-B/M portfolios. In untabulated analyses, we add 76 or 186 industry portfolios based on the MSCI's Global Industry Classification Standard six-digit or eight-digit industries. GICS is a more updated industry classification than the SIC standard used to form Fama French industries.

We continue to find that both TFP and wealth betas are positively associated with cross-sectional stock returns, and the two-beta model's adjusted R^2 is significantly greater than the index beta model's adjusted R^2(p-value$<$0.01). At the industry level, we also continue to find a negative correlation between TFP and wealth betas. For example, copper, electronic components, software, airports, marine ports, highways, and railroads have high TFP betas and low wealth betas, while thrifts and mortgage finance, leisure facilities, and consumable fuel industries have a high wealth betas and low TFP betas. The negative correlation between TFP and wealth betas is more pronounced when using 186 industry portfolios than 76.

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We do not estimate beta directly at the firm level because of the substantial bias imposed by small firms and lack of time-series data for young firms. For example, Lewellen, Nagel, and Shanken (2010, page 182) state: "... an alternative would be to use individual stocks, but errors-in-variables problems could make this impractical, or statistically b[i]ased portfolios, such as those described by Ahn, Conrad and Dittmar, 2009)."[22] Therefore, we estimate earnings beta at the portfolio level then assign those estimates to firms based on the firms' time-varying characteristics.

7.10 Quarterly Analysis

To address a concern about the low (annual) frequency of re-estimating earnings betas, we conduct a quarterly analysis. While this increases the number of earnings observations, quarterly earnings are noisier than annual and subject to seasonality (Kothari, Shanken, and Sloan 1995). The quarterly series is from 1976Q1 to 2017Q4. 1976 is the first time quarterly financial reporting was mandatory. 2017 is the end of the quarterly TFP series estimated by Fernald (2012). Our sample includes US common stocks (CRSP share code: 10 or 11) and is restricted to firms whose fiscal quarters finish at the end of March, June, September, or December. All sample firms are sorted into 81 portfolios each quarter based on the most recent annual B/M, firm size, and asset growth, and the prior q-2 quarter's return momentum MMT, where q is the portfolio formation quarter. We run two quarterly time-series regressions for each portfolio to estimate the three earnings betas (index, wealth and TFP). Each firm's characteristics (B/M ratio, market capitalization, and asset growth) from the same quarter of the previous year or the q-2 quarter's return momentum MMT determine its assignment to a portfolio in quarter q and hence its beta estimates. In untabulated quarterly cross-sectional pricing tests, we find that both TFP and wealth betas are positively associated with cross-sectional stock returns, and the two-beta model's adjusted R^2 is significantly greater than the index beta model's adjusted R^2 (p-value$<$0.01). In addition, we continue to find a negative correlation between TFP and wealth betas.

[22] Specifically, Ahn et al. (2009, Figure 3) show that Sharpe ratio bias increases with the number of basis asset portfolios.

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7.11 Portfolio Pricing Tests

We next conduct portfolio-level pricing tests. We use an annual series of aggregate earnings and changes in total factor productivity and aggregate wealth from 1964-2018 to estimate three earnings betas. A firm's characteristics (B/M ratio, cumulative return, market capitalization, and asset growth) at the end of year t-1 determine its assignment to a portfolio and hence its beta estimates. Twelve subsequent monthly returns are merged with these betas, starting in the seventh month after portfolio formation. We calculate the equal-weighted or value-weighted (based on beginning market capitalization) portfolio monthly or annual returns and run monthly or annual cross-sectional pricing tests, correspondingly. In untabulated results across the four different specifications, we consistently find that both TFP and wealth betas are positively associated with cross-sectional stock returns, and the two-beta model's adjusted R^2 is significantly greater than the index beta model's adjusted R^2 (p-value$<$0.01). In sum, our inferences are not altered due to different pricing test specifications.

8 Conclusion

We use macroeconomic indicators rather than indexes of public-firm earnings to measure aggregate risk on the grounds that equity claims on public firms reflect a minor and unrepresentative subset of assets generating aggregate risk. This leads us to use realized accounting earnings changes rather than stock returns to estimate firm-level systematic risk, on the grounds that earnings and macroeconomic indicators are primarily actual annual outcomes that are well aligned in time, and thus offer the prospect of capturing the contemporaneous comovements in earnings that underlie systematic risk. Stock returns, however, incorporate changes in expected future outcomes. In using earnings when estimating systematic risk we build on an extensive literature including Ball and Brown (1969), Beaver, Kettler, and Scholes (1970), Rosenberg and McKibben (1973), Beaver and Manegold (1975), Fama and French (1995), Ball, Sadka, and Sadka (2009), Da and Warachka (2009), Ellahie (2020), and

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Penman and Zhang (2020).

We employ two measures of firm-level systematic risk to capture the sensitivity of accounting earnings to macroeconomic indicators of aggregate economy-wide shocks. The macroeconomic indicators studied are aggregate supply shocks (instrumented by changes in Total Factor Productivity) and aggregate demand shocks (instrumented by changes in total household wealth). Together these provide a parsimonious model of the two fundamental forces affecting aggregate risk. We do not propose that our instruments for aggregate supply and demand shocks are perfect, or close thereto; rather, we offer them as capturing outcomes on a wider range of economic activity than the "market" indexes typically used in prior literature, and hence as providing potentially better measures of aggregate asset risk and providing some insights into its origins.

We find that firms' earnings-based supply and demand betas are negatively correlated. This indicates that firms affected more by aggregate demand shocks are affected less by aggregate supply shocks, and vice versa. In Fama and MacBeth (1973) regressions, earnings-based supply and demand betas both carry an economically and statistically significant positive risk premium. We also find that firm-level sensitivities (betas) to aggregate supply and demand shocks are correlated with four "empirical" firm-level characteristics that help explain the cross-section of stock returns. After controlling for asset supply and demand betas, the collective explanatory power of these firm characteristics with respect to the cross-section of stock returns declines by one quarter. These findings provide an important clue as to why these firm characteristics are priced.

These results are based on instruments for aggregate supply and demand shocks that inevitably are imperfect. That said, we view the instruments as superior to a specification in which total public-firm equity proxies for aggregate outcomes. We also view the results as further validating the use of accounting earnings to estimate firm-level systematic risk.

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Table 1: Summary Statistics of Aggregate Variables

Our sample includes US common stocks (CRSP share code: 10 or 11). Aggregate earnings changes is the sum of earnings in year t from all sample firms minus the sum of earnings from year t-1 divided by the sum of market capitalization in the end of year t-2. Earnings is operating income after depreciation (Compustat: OIADP) minus preferred stock dividends (PDVC). If OIADP is missing, we use IB (income before extraordinary items) instead. ΔTFP is $\Delta \ln Y - \alpha \Delta \ln K - (1-\alpha)\Delta \ln L$ (i.e., change in output ΔY less change in the contribution of capital K and labor L) estimated by Fernald (2012). ΔWealth is the percentage change in aggregate wealth, which is household total net worth (Fed: Z1/Z1/FL152090005) minus corporate equities (Z1/Z1/FL153064105) and indirectly held corporate equities (Z1/Z1/FL153064175). To adjust for high serial correlation, we use the first difference of the annual change in wealth. Annual series are from 1964 to 2018 (N=55). Panel A presents distributions of aggregate variables used in the analysis. Panel B shows their serial correlations after adjustments. Panel C reports cross correlation matrices (Right: Pearson; Left: Spearman).

Panel A: Distribution of annual aggregate variables

	Mean	25th	50th	75th	Std Dev
Aggregate earnings changes	0.025	0.014	0.025	0.035	0.020
ΔTFP	0.009	0.000	0.010	0.020	0.016
ΔWealth	0.000	-0.017	0.002	0.015	0.027

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Panel B: Serial correlation of annual aggregate variables					
	1st lag	2nd lag	3rd lag	4th lag	5th lag
Aggregate earnings changes	0.14	-0.32	-0.23	0.07	-0.04
ΔTFP	0.11	-0.09	-0.19	0.00	0.04
ΔWealth	0.11	-0.13	-0.22	-0.25	-0.07

Panel C: Cross correlation of annual aggregate variables			
	$\Delta Earnings_{MKT}$	ΔTFP	ΔWealth
Aggregate earnings changes		0.44	0.35
ΔTFP	0.38		0.49
ΔWealth	0.25	0.50	

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Table 2: Aggregate Time-Series Regressions

This table reports results from aggregate time series regressions: $\Delta W_{t,t+k} = \alpha_0 + \alpha_1 \Delta Z_t + \varepsilon_{\alpha,t}$ where $\Delta W_{t,t+k}$ is the cumulated growth of W from year t to year $t+k$ and ΔZ_t is the annual growth of Z in year t. In Panel A, $k = 0$, ΔW is aggregate earnings changes and ΔZ is ΔTFP, ΔWealth, or both. In Panel B, ΔW is aggregate earnings changes and ΔZ is aggregate stock return measured as value weighted returns from May of year t to April of year $t+1$ based on market capitalization from year $t-1$ and adjusted for delisted returns. In Panel C, ΔW is the adjusted GDP growth based on an AR(1) process and ΔZ is aggregate stock return measured as value weighted returns from Jan. of year t to Dec. of year t based on market capitalization from year $t-1$ and adjusted for delisted returns (calendar return). In Panel D, ΔW is the adjusted consumption growth based on an AR(1) process and ΔZ is aggregate calendar stock return. In Panel E, ΔW is total factor productivity growth and ΔZ is aggregate calendar stock return. In Panel F, ΔW is wealth growth and ΔZ is aggregate calendar stock return. In Panel G, ΔW is total factor productivity growth and ΔZ is aggregate earnings changes. In Panel H, ΔW is wealth growth and ΔZ is aggregate earnings changes. In the lead-lag analysis, sample firms are restricted to those with fiscal year ended in December. To adjust for high serial correlations, we use the first difference of wealth growth. Annual series are from 1964 to 2018 (N=55). The t-statistics and p-values are based on heteroscedasticity-consistent standard errors.

Panel A: $\Delta Earnings_{MKT,t} = \alpha_{A0} + \alpha_{A1} \Delta TFP_t / \Delta Wealth_t + \varepsilon_{\alpha A,t}$

Intercept	t	ΔTFP	t	$\Delta Wealth$	t	Adj. R^2
0.020	6.53	0.551	2.88			0.18
0.025	10.08			0.259	1.91	0.06
0.021	6.85	0.441	2.13	0.132	1.00	0.19

Panel B: $\Delta Earnings_{MKT,t,t+k} = \alpha_{B0} + \alpha_{B1} Return_{MKT,t} + \varepsilon_{\alpha B,t}$

$k =$	$Return_{MKT}$	p-value	Adj. R^2
0	0.011	0.65	-0.01
1	0.060	0.06	0.07
2	0.059	0.02	0.05
3	0.028	0.18	-0.01
4	0.022	0.41	-0.01
5	-0.010	0.77	-0.02

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Panel C: $\Delta GDP_{t,t+k} = \alpha_{C0} + \alpha_{C1} Return_{MKT,t} + \varepsilon_{\alpha C,t}$			
$k =$	$Return_{MKT}$	p-value	Adj. R^2
0	0.006	0.75	-0.02
1	0.075	<0.01	0.19
2	0.051	0.05	0.05
3	0.041	0.13	0.02
4	0.050	0.04	0.04
5	0.037	0.09	0.00

Panel D: $\Delta Consumption_{t,t+k} = \alpha_{D0} + \alpha_{D1} Return_{MKT,t} + \varepsilon_{\alpha D,t}$			
$k =$	$Return_{MKT}$	p-value	Adj. R^2
0	0.011	0.30	0.00
1	0.011	0.47	-0.01
2	0.026	0.22	0.01
3	0.034	0.15	0.02
4	0.043	0.08	0.04
5	0.047	0.12	0.02

Panel E: $\Delta TFP_{t,t+k} = \alpha_{E0} + \alpha_{E1} Return_{MKT,t} + \varepsilon_{\alpha E,t}$			
$k =$	$Return_{MKT}$	p-value	Adj. R^2
0	0.008	0.63	-0.01
1	0.051	0.01	0.12
2	0.019	0.39	-0.01
3	0.006	0.80	-0.02
4	0.004	0.87	-0.02
5	0.009	0.75	-0.02

Panel F: $\Delta Wealth_{t,t+k} = \alpha_{F0} + \alpha_{F1} Return_{MKT,t} + \varepsilon_{\alpha F,t}$			
$k =$	$Return_{MKT}$	p-value	Adj. R^2
0	0.036	0.09	0.03
1	0.058	0.02	0.04
2	-0.001	0.98	-0.02
3	-0.007	0.82	-0.02
4	-0.019	0.64	-0.02
5	-0.020	0.65	-0.02

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Panel G: $\Delta TFP_{t,t+k} = \alpha_{G0} + \alpha_{G1}\Delta Earnings_{MKT,t} + \varepsilon_{\alpha G,t}$			
$k =$	$\Delta Earnings_{MKT}$	p-value	Adj. R^2
0	0.318	<0.01	0.16
1	0.279	0.08	0.04
2	-0.009	0.96	-0.02
3	-0.257	0.30	0.01
4	-0.172	0.48	-0.01
5	-0.310	0.19	0.00
Panel H: $\Delta Wealth_{t,t+k} = \alpha_{H0} + \alpha_{H1}\Delta Earnings_{MKT,t} + \varepsilon_{\alpha H,t}$			
$k =$	$\Delta Earnings_{MKT}$	p-value	Adj. R^2
0	0.477	0.02	0.13
1	0.275	0.28	0.00
2	-0.275	0.46	-0.01
3	-0.534	0.22	0.03
4	-0.960	0.01	0.12
5	-0.505	0.12	0.02

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Table 3: Book-to-market Portfolios and Supply and Demand Betas
This table reports results from book-to-market portfolio time series regressions: $\Delta Earning_{t,p} = \beta_{0p} + \beta_{1p}\Delta Z_t + \varepsilon_{\beta,t,p}$ where $Z = TFP, Wealth, or\ both$.Sample firms are sorted into five portfolios based on their lagged B/M ratios. B/M is book equity at the fiscal period end divided by market equity (CRSP: PRC, SHROUT) at the end of December. Portfolio earnings changes is the sum of earnings in year t from all firms in a portfolio minus the sum of earnings from year t-1 divided by the sum of market capitalization in the end of year t-2. Earnings is operating income after depreciation (Compustat: OIADP) minus preferred stock dividends (PDVC). If OIADP is missing, we use IB (income before extraordinary items) instead. ΔTFP is $\Delta \ln Y - \alpha\Delta \ln K - (1-\alpha)\Delta \ln L$ (i.e., change in output ΔY less change in the contribution of capital K and labor L) estimated by Fernald (2012). ΔWealth is the percentage change in aggregate wealth. Aggregate wealth is household total net worth (Fed: Z1/Z1/FL152090005) minus corporate equities (Z1/Z1/FL153064105) and indirectly held corporate equities (Z1/Z1/FL153064175). To adjust for high serial correlation, we use the first difference of annual change in wealth. Annual series are from 1964 to 2018 (N=55). Firm-level earnings are winsorized at $+/-1\%$. The t-statistics are based on heteroscedasticity-consistent standard errors.

Annual book-to-market portfolio time-series regression							
	Intercept	t	ΔTFP	t	ΔWealth	t	Adj. R^2
High B/M	0.011	1.91	1.352	3.55			0.24
4	0.005	1.41	0.602	2.79			0.20
3	0.008	3.36	0.403	2.84			0.18
2	0.008	4.13	0.221	1.89			0.10
Low B/M	0.012	7.95	0.077	0.87			0.00
High B/M	0.023	4.33			0.550	3.16	0.10
4	0.010	4.02			0.263	2.25	0.10
3	0.011	5.96			0.129	1.84	0.04
2	0.010	7.51			0.087	1.68	0.03
Low B/M	0.012	10.29			0.031	0.76	-0.01
High B/M	0.013	2.08	1.174	2.57	0.214	1.14	0.24
4	0.006	1.55	0.503	1.94	0.119	0.96	0.21
3	0.008	3.12	0.388	2.25	0.018	0.24	0.16
2	0.009	3.85	0.195	1.40	0.031	0.53	0.08
Low B/M	0.012	7.12	0.067	0.64	0.012	0.24	-0.02

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Table 4: Return Momentum Portfolios and Supply and Demand Betas
This table reports results from return momentum portfolio time series regressions: $\Delta Earning_{t,p} = \beta_{0p} + \beta_{1p}\Delta Z_t + \varepsilon_{\beta,t,p}$ where $Z = TFP, Wealth, or\ both$. Sample firms are sorted into five portfolios based on cumulated continuously compounded stock returns from $j-1$ to $j-12$ where j is the portfolio formation month. Portfolio earnings changes is the sum of earnings in year t from all firms in a portfolio minus the sum of earnings from year t-1 divided by the sum of market capitalization in the end of year t-2. Earnings is operating income after depreciation (Compustat: OIADP) minus preferred stock dividends (PDVC). If OIADP is missing, we use IB (income before extraordinary items) instead. ΔTFP is $\Delta \ln Y - \alpha \Delta \ln K - (1-\alpha)\Delta \ln L$ (i.e., change in output ΔY less change in the contribution of capital K and labor L) estimated by Fernald (2012). ΔWealth is the percentage change in aggregate wealth. Aggregate wealth is household total net worth (Fed: Z1/Z1/FL152090005) minus corporate equities (Z1/Z1/FL153064105) and indirectly held corporate equities (Z1/Z1/FL153064175). To adjust for high serial correlation, we use the first difference of annual change in wealth. Annual series are from 1964 to 2018 (N=55). Firm-level earnings are winsorized at $+/-1\%$. The t-statistics are based on heteroscedasticity-consistent standard errors.

Annual momentum portfolio time-series regression							
	Intercept	t	ΔTFP	t	ΔWealth	t	Adj. R^2
High momentum	0.039	8.30	0.883	2.77			0.11
4	0.021	7.56	0.537	2.55			0.12
3	0.013	6.09	0.404	2.99			0.15
2	0.007	3.18	0.448	3.91			0.20
Low momentum	0.002	0.69	0.444	3.30			0.09
High momentum	0.047	9.22			0.484	1.73	0.09
4	0.026	8.77			0.292	1.79	0.10
3	0.016	8.43			0.236	2.39	0.15
2	0.011	6.02			0.226	2.72	0.15
Low momentum	0.006	2.33			0.242	2.22	0.08
High momentum	0.041	7.28	0.631	1.91	0.303	1.04	0.12
4	0.023	6.86	0.387	1.74	0.181	1.07	0.13
3	0.014	5.92	0.273	1.89	0.158	1.50	0.19
2	0.008	3.59	0.340	2.53	0.129	1.50	0.23
Low momentum	0.003	1.22	0.319	2.65	0.150	1.46	0.10

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Table 5: Firm Size Portfolios and Supply and Demand Betas

This table reports results from size portfolio time series regression: $\Delta Earning_{t,p} = \beta_{0p} + \beta_{1p}\Delta Z_t + \varepsilon_{\beta,t,p}$ where $Z = TFP, Wealth, or\ both.$ Sample firms are sorted into five portfolios based on their lagged market capitalization in the end of year t-1. Portfolio earnings changes is the sum of earnings in year t from all firms in a portfolio minus the sum of earnings from year t-1 divided by the sum of market capitalization in the end of year t-2. Earnings is operating income after depreciation (Compustat: OIADP) minus preferred stock dividends (PDVC). If OIADP is missing, we use IB (income before extraordinary items) instead. ΔTFP is $\Delta \ln Y - \alpha\Delta \ln K - (1-\alpha)\Delta \ln L$ (i.e., change in output ΔY less change in the contribution of capital K and labor L) estimated by Fernald (2012). ΔWealth is the percentage change in aggregate wealth. Aggregate wealth is household total net worth (Fed: Z1/Z1/FL152090005) minus corporate equities (Z1/Z1/FL153064105) and indirectly held corporate equities (Z1/Z1/FL153064175). To adjust for high serial correlation, we use the first difference of annual change in wealth. Annual series are from 1964 to 2018 (N=55). Firm-level earnings are winsorized at $+/-1\%$. The t-statistics are based on heteroscedasticity-consistent standard errors.

Annual size portfolio time-series regression							
	Intercept	t	ΔTFP	t	ΔWealth	t	Adj. R^2
Large size	0.007	4.05	0.294	2.63			0.17
4	0.014	3.65	0.692	3.06			0.20
3	0.018	4.80	0.784	3.58			0.21
2	0.021	4.14	1.125	3.47			0.20
Small size	0.034	5.06	1.267	2.98			0.16
Large size	0.010	7.09			0.128	2.06	0.08
4	0.020	6.89			0.345	3.29	0.14
3	0.025	7.86			0.380	3.75	0.14
2	0.032	6.58			0.529	3.98	0.12
Small size	0.046	7.55			0.585	3.29	0.09
Large size	0.008	4.03	0.246	1.95	0.058	0.96	0.17
4	0.015	3.76	0.532	1.94	0.193	1.65	0.22
3	0.020	5.06	0.614	2.38	0.205	1.94	0.24
2	0.023	4.66	0.900	2.34	0.271	1.86	0.21
Small size	0.036	5.29	1.025	1.99	0.291	1.41	0.17

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Table 6: Asset Growth Portfolios and Supply and Demand Betas

This table reports results from asset growth portfolio time series regression: $\Delta Earning_{t,p} = \beta_{0p} + \beta_{1p}\Delta Z_t + \varepsilon_{\beta,t,p}$ where $Z = TFP, Wealth, or\ both$. Sample firms are sorted into five portfolios based on their lagged asset growth in year t-1. Asset growth is the change in the natural log of book value of assets per split-adjusted share. Portfolio earnings changes is the sum of earnings in year t from all firms in a portfolio minus the sum of earnings from year t-1 divided by the sum of market capitalization in the end of year t-2. Earnings is operating income after depreciation (Compustat: OIADP) minus preferred stock dividends (PDVC). If OIADP is missing, we use IB (income before extraordinary items) instead. ΔTFP is $\Delta \ln Y - \alpha\Delta \ln K - (1-\alpha)\Delta \ln L$ (i.e., change in output ΔY less change in the contribution of capital K and labor L) estimated by Fernald (2012). ΔWealth is the percentage change in aggregate wealth. Aggregate wealth is household total net worth (Fed: Z1/Z1/FL152090005) minus corporate equities (Z1/Z1/FL153064105) and indirectly held corporate equities (Z1/Z1/FL153064175). To adjust for high serial correlation, we use the first difference of annual change in wealth. Annual series are from 1964 to 2018 (N=55). Firm-level earnings are winsorized at $+/-1\%$. The t-statistics are based on heteroscedasticity-consistent standard errors.

Annual asset growth portfolio time-series regression							
	Intercept	t	ΔTFP	t	ΔWealth	t	Adj. R^2
High growth	0.012	4.46	0.227	1.65			0.04
4	0.005	2.68	0.228	2.32			0.14
3	0.004	2.20	0.321	3.18			0.24
2	0.006	2.34	0.479	2.37			0.19
Low growth	0.019	4.90	0.813	2.66			0.15
High growth	0.014	7.11			0.072	1.09	0.00
4	0.007	5.66			0.100	2.07	0.07
3	0.007	5.20			0.134	2.83	0.11
2	0.011	5.17			0.240	2.36	0.13
Low growth	0.026	6.63			0.423	3.29	0.12
High growth	0.012	4.03	0.009	0.12	0.220	1.28	0.02
4	0.005	2.72	0.045	0.91	0.190	1.63	0.13
3	0.004	2.31	0.055	1.19	0.275	2.29	0.25
2	0.007	2.53	0.136	1.37	0.366	1.70	0.21
Low growth	0.021	5.42	0.250	1.64	0.605	1.68	0.17

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Table 7: Summary Statistics of Earnings Betas

This table presents summary statistics of three estimated earnings betas: Index beta $\widehat{\beta}_{MKT,p}$, TFP beta $\widehat{\beta}_{TFP,p}$, and wealth beta $\widehat{\beta}_{Wealth,p}$. All sample firms are sorted into 81 portfolios based on lagged B/M, return momentum MMT, firm size, and asset growth ATG (i.e., 3 B/M terciles $\times$ 3 return momentum terciles $\times$ 3 size terciles $\times$ 3 asset growth terciles). Then we run two time-series regressions for each portfolio: 1) $\Delta Earnings_{t,p} = \gamma_{0p} + \beta_{MKT,p}\Delta Earnings_{MKT,t} + \varepsilon_{\gamma,t,p}$ and 2) $\Delta Earnings_{t,p} = \delta_{0p} + \beta_{Wealth,p}\Delta Wealth_t + \beta_{TFP,p}\Delta TFP_t + \varepsilon_{\delta,t,p}$. $\Delta Earnings_{MKT,t}$ is aggregate earnings changes measured as the sum of earnings in year t from all sample firms minus the sum of earnings from year t-1 divided by the sum of market capitalization in the end of year t-2. ΔTFP is $\Delta \ln Y - \alpha \Delta \ln K - (1-\alpha)\Delta \ln L$ (i.e., change in output ΔY less change in the contribution of capital K and labor L) estimated by Fernald (2012). $\Delta Wealth$ is the percentage change in aggregate wealth. Aggregate wealth is household total net worth (Fed: Z1/Z1/FL152090005) minus corporate equities (Z1/Z1/FL153064105) and indirectly held corporate equities (Z1/Z1/FL153064175). To adjust for high serial correlation, we use the first difference of annual change in wealth. Annual series are from 1964 to 2018 (N=55). Firm-level earnings are winsorized at $+/-1\%$. Panel A presents distributions of estimated betas. Panel B reports cross correlation matrices (Right: Pearson; Left: Spearman). Panel C presents the percentage of firms remaining in the same portfolio in year t+k where year t is the portfolio formation year.

Panel A: Distribution of earnings betas (N=81)

	Mean	25th	50th	75th	Std Dev
Index beta	1.92	1.17	1.58	2.59	1.23
TFP beta	0.60	0.20	0.52	0.90	0.64
Wealth beta	0.16	0.05	0.14	0.26	0.28

Panel B: Cross correlation of earnings betas and firm characteristics

	$\widehat{\beta}_{MKT}$	$\widehat{\beta}_{TFP}$	$\widehat{\beta}_{Wealth}$	$\log(B/M)$	MMT	$\log(SIZE)$	ATG
Index beta		0.04	0.40	0.18	0.06	-0.08	-0.28
TFP beta	0.43		-0.46	0.23	-0.10	-0.10	-0.03
Wealth beta	0.35	-0.23		0.10	0.01	-0.03	-0.34
$\log(B/M)$	0.42	0.58	0.26		-0.13	-0.04	-0.05
MMT	0.08	-0.14	-0.01	-0.27		0.01	0.21
$\log(SIZE)$	-0.32	-0.47	-0.22	-0.38	0.12		0.02
ATG	-0.33	-0.17	-0.48	-0.17	0.21	0.15	

Panel C: Serial correlation of firm-year earnings betas

	1st lag	2nd lag	3rd lag	4th lag	5th lag
Index beta	0.66	0.30	0.27	0.27	0.25
TFP beta	0.70	0.38	0.35	0.35	0.34
Wealth beta	0.59	0.15	0.12	0.13	0.11

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Table 8: Cross-sectional Pricing Tests

In this table, we report the average monthly intercept and slope coefficient, along with Fama and MacBeth (1973) t-statistics and the average regression adjusted R^2. A firm's characteristics (B/M ratio, cumulative return, market capitalization, and asset growth) at the end of year $t-1$ determine its assignment to a portfolio and hence its beta estimates. Twelve subsequent monthly returns, starting in the seventh month after portfolio formation, are regressed individually on the earnings betas. In each month, we estimate the following cross-sectional regressions: Model (1) $RET_{m,i} = \eta_{0m} + \eta_{1m}\widehat{\beta}_{MKT,t,i} + \varepsilon_{\eta,m,i}$, Model (2) $RET_{m,i} = \lambda_{0m} + \lambda_{1m}\widehat{\beta}_{TFP,t,i} + \lambda_{2m}\widehat{\beta}_{Wealth,t,i} + \varepsilon_{\lambda,m,i}$, Model (3) $RET_{m,i} = \theta_{0m} + \theta_{1m}log(B/M)_{t,i} + \theta_{2m}MMT_{t,i} + \theta_{3m}log(SIZE)_{t,i} + \theta_{4m}ATG_{t,i} + \varepsilon_{\theta,m,i}$, and Model (4) $RET_{m,i} = \nu_{0m} + \nu_{1m}\,\widehat{\varepsilon}_{log(B/M),t,i} + \nu_{2m}\,\widehat{\varepsilon}_{MMT,t,i} + \nu_{3m}\,\widehat{\varepsilon}_{log(SIZE),t,i} + \nu_{4m}\,\widehat{\varepsilon}_{ATG,t,i} + \varepsilon_{\nu,m,i}$ where $\widehat{\varepsilon}_{K,t,i}$ is the residual from regressing a firm characteristic K on $\widehat{\beta}_{TFP}$ and $\widehat{\beta}_{Wealth}$ in each year; for example, $log(B/M)_{t,i} = \omega_{0t} + \omega_{1t}\widehat{\beta}_{TFP,t,i} + \omega_{2t}\widehat{\beta}_{Wealth,t,i} + \varepsilon_{\omega,t,i}$. The t-statistics are based on heteroscedasticity-consistent standard errors. We also report t-statistics based on the Newey-West (1994) standard errors adjusting for twelve-month lags (t_{NW}). In addition, we accumulate twelve-month returns and conduct an annual return pricing test as a robustness check (t_{Ann}, $Adj.R^2_{Ann}$). Lastly, GLS adjusted R^2 is also reported ($Adj.R^2_{GLS}$), following the suggestion by Lewellen, Nagel, and Shanken (2010).

Panel A: Cross-sectional pricing test on earnings betas

Model (1): $RET_{m/t,i} = \eta_{0m/t} + \eta_{1m/t}\widehat{\beta}_{MKT,t,i} + \varepsilon_{\eta,m/t,i}$

Intercept	t	$\widehat{\beta}_{MKT}$	t	t_{NW}	t_{Ann}	$Adj.R^2$	$Adj.R^2_{GLS}$	$Adj.R^2_{Ann}$
0.86%	4.23	0.20%	5.66	4.59	2.24	0.32%	3.60%	0.53%

Model (2): $RET_{m/t,i} = \lambda_{0m/t} + \lambda_{1m/t}\widehat{\beta}_{TFP,t,i} + \lambda_{2m/t}\widehat{\beta}_{Wealth,t,i} + \varepsilon_{\lambda,m/t,i}$

Intercept	t	$\widehat{\beta}_{TFP}$	t	t_{NW}	t_{Ann}	$\widehat{\beta}_{Wealth}$	t	t_{NW}	t_{Ann}	$Adj.R^2$	$Adj.R^2_{GLS}$	$Adj.R^2_{Ann}$
0.82%	3.87	0.41%	5.47	4.66	4.26	0.88%	7.30	5.99	1.90	0.63%	2.89%	1.03%

Panel B: Cross-sectional pricing test on firm characteristics									
Model:	(3)					(4)			
Dependent variable:	$RET_{m,i}$					$RET_{m,i}$			
	$Coeff.$	t	t_{NW}	t_{Ann}		$Coeff.$	t	t_{NW}	t_{Ann}
$Intercept$	1.73%	5.66	4.70	3.68	$Intercept$	1.23%	5.61	5.28	4.51
$log(B/M)$	0.21%	4.07	3.38	1.08	$\varepsilon_{log(B/M)}$	0.21%	3.88	3.23	0.10
MMT	-0.10%	-1.29	-1.31	0.82	ε_{MMT}	-0.11%	-1.40	-1.41	-0.10
$log(SIZE)$	-0.09%	-2.56	-2.27	-1.75	$\varepsilon_{log(SIZE)}$	-0.09%	-2.68	-2.41	-1.26
ATG	-0.62%	-6.09	-5.22	-1.64	ε_{ATG}	-0.56%	-5.35	-4.89	-2.25
$Adj.R^2$	2.42%				$Adj.R^2$	1.83%			
$Adj.R^2_{GLS}$	7.99%				$Adj.R^2_{GLS}$	5.74%			
$Adj.R^2_{Ann}$	3.71%				$Adj.R^2_{Ann}$	2.76%			

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Table 9: Effect of Firm-level TFP

In this table, we report results from Fama and MacBeth regressions incorporating firm-level TFP in the Model (2) regression reported in Table 8 Panel A. The sample includes firms for which we have TFP data from İmrohoroğlu and Tüzel (2014), excluding financials and utilities, ended in 2013, and requiring a positive number of employees and a positive value for other five variables in Compustat. TFP is from year $t-1$. Also, a firm's characteristics (B/M ratio, cumulative return, market capitalization, and asset growth) at the end of year $t-1$ determine its assignment to a portfolio and hence its beta estimates. Twelve subsequent monthly returns, starting in the seventh month after portfolio formation, are regressed individually on the earnings betas and/or TFP. We report the average monthly intercept and slope coefficient, along with Fama and MacBeth (1973) t-statistics and the average regression adjusted R^2.

Model (1): $RET_{m,i} = \lambda_{0m} + \lambda_{1m}\widehat{\beta}_{TFP,t,i} + \lambda_{2m}\widehat{\beta}_{Wealth,t,i} + \varepsilon_{\lambda,m,i}$

Intercept	t	$\widehat{\beta}_{TFP}$	t	$\widehat{\beta}_{Wealth}$	t			$Adj.R^2$
0.92%	3.94	0.40%	4.78	0.71%	5.46			0.85%

Model (2): $RET_{m,i} = \lambda_{3m} + \lambda_{4m}TFP_{t-1,i} + \varepsilon_{\lambda,m,i}$

Intercept	t					TFP	t	$Adj.R^2$
1.23%	5.34					-0.18%	-1.43	0.52%

Model (3): $RET_{m,i} = \lambda_{5m} + \lambda_{6m}\widehat{\beta}_{TFP,t,i} + \lambda_{7m}TFP_{t-1,i} + \varepsilon_{\lambda,m,i}$

Intercept	t	$\widehat{\beta}_{TFP}$	t			TFP	t	$Adj.R^2$
1.07%	4.59	0.37%	5.42			0.00%	-0.04	0.95%

Model (4): $RET_{m,i} = \lambda_{8m} + \lambda_{9m}\widehat{\beta}_{TFP,t,i} + \lambda_{10m}\widehat{\beta}_{Wealth,t,i} + \lambda_{11m}TFP_{t-1,i} + \varepsilon_{\lambda,m,i}$

Intercept	t	$\widehat{\beta}_{TFP}$	t	$\widehat{\beta}_{Wealth}$	t	TFP	t	$Adj.R^2$
0.93%	4.02	0.41%	5.67	0.75%	6.69	0.09%	0.96	1.12%

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